

The Closure Law of a Finite Index

A finite set of tuples over chains determines, from its own extension alone, a smallest sublattice containing it — the staircase cut by its pairwise monotone bounds — and the number of cells that sublattice adds is a measurable defect of the index.

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Abstract

Let X be a finite non-empty set of d -tuples whose coordinates take values in finite chains. Two things can be read off X without any outside knowledge: the set of values it realises at each coordinate, and, for each ordered pair of coordinates (i, j) , the monotone bound $\phi_{ij}(a) = \max\{y_i : y \in X, y_j \leq a\}$. The cells admitted by those two readings form a set $\mathcal{R}(X) \supseteq X$, and the difference $E(X) = |\mathcal{R}(X)| - |X|$ is the index's **closure defect**. This paper establishes what $\mathcal{R}$ is. It is a closure operator — extensive, monotone and idempotent — whose image is always a sublattice of the product of chains, and it is the *smallest* such sublattice containing X : $\mathcal{R}(X) = \langle X \rangle$. Consequently X is closed under coordinatewise minimum and maximum exactly when $E(X) = 0$, which identifies a combinatorial defect with a lattice-theoretic property. The identification rests on a classical theorem — a sublattice of a finite product of lattices is determined by its two-fold projections, Baker and Pixley's interpolation theorem for algebras with a majority term, the projection-determination property that Bergman named — and the paper says exactly which step is imported and which is proved here. Six consequences follow and are developed in full: the boundary functions are the pointwise least isotone bound system representing X , so a closed index carries a canonical system of inequalities; the closed subsets of a fixed ambient form a Moore family whose own defect is exactly $2^n - |\text{Cl}(U)|$ for an ambient of n cells, so the family of closed indexes is itself maximally open, its own closure being the entire power set; projections of closed sets are closed and the converse fails; adjoining a derived coordinate never repairs closure, the graph of a map being closed precisely when the map is a lattice homomorphism, which excludes every coordinate difference; the bands $|a - b| \leq k$ are sublattices while the triangle region $|a - b| \leq c \leq a + b$ is join-closed and meet-broken, with explicit witnesses; and the minimum generating set of a closed index is an exact minimum set cover, with a closed-form law for the ordered simplex — the lattice of Young diagrams in a box — and, for the full box, Czédli's law for the generating number of a direct power of a chain, given here a second proof from Bollobás's set-pair inequality and used to refute the linear law that a small sample suggests. Every decidable claim is machine-checked by an SMT solver over every subset of a named finite box, or decided exhaustively over a stated finite family: 53 machine-checked obligations, 47 exhaustive families, 6 refutations — five by explicit witness and one by solver — and 8 results taken from the literature.

§0 · The result

A finite index has a closure, the closure is its lattice hull, and the gap between them is a number worth measuring.

An index is a finite table: cells over d coordinates, each coordinate taking values in a chain. The table does not say what else could exist. But it does not say nothing, either. Two features of a table are visible without any outside knowledge — which values each coordinate takes, and how far one coordinate reaches given a bound on another — and together they admit a determinate set of further cells. Writing $\mathcal{R}(X)$ for that set and $E(X) = |\mathcal{R}(X)| - |X|$ for the count of cells it adds, the paper's first result (Theorem 1, Theorem 2, Theorem 3) is that this apparently ad hoc reconstruction is exactly the lattice hull:

$\mathcal{R}$ is a closure operator, $\mathcal{R}(X)$ is the smallest sublattice of the ambient product containing X , and $E(X) = 0$ if and only if X is closed under coordinatewise minimum and maximum.

That is the closure law. It converts a question about a table — *what does this index leave unstated?* — into a question about a lattice, and it makes E a computable property of a layout rather than of its subject matter. Three illustrations run through §2. A common year of the Gregorian calendar, indexed by (month, day), has $E = 7$: the seven cells are 29, 30 and 31 February and 31 April, June, September, November, and they are the content of the rhyme that has to be taught alongside the calendar. The periodic table as usually drawn, eighteen columns with the f block detached, has $E = 36$, and moving one element — helium, from group 18 to group 2 — takes that to 20. E is a property of the coordinatisation, not of the elements.

What is proved, and what is imported. The closure axioms (Theorem 1), the sublattice property of the image (Lemma 4), the characterisation of closed sets (Theorem 3) and all six consequences are proved here in full. The one imported step is the lift of the hull identity from $d = 2$ to $d \geq 3$. At $d = 2$ this paper proves it outright, by an explicit four-witness construction (Lemma 5). Above $d = 2$ it follows from the determination of a sublattice of a finite product by its two-fold projections, which is Baker and Pixley’s (1975) interpolation theorem for any algebra with a majority term, of which the lattice median is one; Bergman (1977) named the property and proved its converse. That step is marked CITED wherever it is used and is not reproved. In the language of constraint satisfaction, $E(X) = 0$ says that X is the solution set of a binary network of staircase constraints — binary-decomposable, in Montanari’s (1974) sense, into monotone constraints — and that network, being path consistent and row convex, is then globally consistent by van Beek and Dechter (1995); §3 says why the decomposition, and not the consistency, is what $E = 0$ names. The constraint class that φ recovers is the staircase, or connected row-convex, class of Deville, Barette and Van Hentenryck (1999). **Freuder (1982) is a different theorem and is not among these:** it gives backtrack-free search on a graph of width w under strong $(w+1)$ -consistency, which is a statement about search and not about decomposability.

What is not established. The hull identity is proved here only at $d = 2$; above that it rests on the cited theorem, and the machine checks cover it over the named boxes and no further. The representation requires each coordinate to be presented as a chain. That is not a requirement that the factors be chains: a finite distributive factor L embeds as a sublattice of $\{0,1\}^J$, J its set of join-irreducibles, one two-element chain per irreducible, by Birkhoff’s representation theorem (Birkhoff 1967). It is the requirement that the presentation not fold two chains into a single non-chain coordinate, because $\mathcal{R}$ reconstructs coordinate by coordinate. Nothing here is a claim about infinite index sets. E is coordinate-relative by construction: it measures a coordinatisation, and a set of composite size can always be relabelled onto a full rectangle, which is closed. The seed laws of §9 are exact for the two families named there and for those only. The full-box law is Czédli’s theorem on generating a direct power of a chain — cited, and given a second proof — and not a result of this paper; in general seed is an instance of minimum set cover, which is NP-complete (Karp 1972), a bound on its difficulty from above and not a proof that it is hard, and no closed form is claimed beyond the two families.

Status words. Six are used and never merged.

status	meaning
PROVED	a proof is written out below and every step is justified
MACHINE-CHECKED	an SMT solver returned unsat on the negation of an obligation whose variables range over <i>every</i> subset of a named finite box — or, for the integer obligations, over every integer — with both guards passed
EXHAUSTIVE	a decision procedure visited every case in a stated finite family
REFUTATION	a claim disproved by an explicit witness re-verified here, or by a solver’s satisfying assignment

status	meaning
CITED	taken from the literature, with the full source
SAMPLED	a seeded pseudorandom sweep of a stated size; used below only for the two soundness guards, never for a result

A machine-checked claim names its box; an exhaustive claim names its family. §10 names all of them.

§1 • Definitions

Throughout, $d \geq 2$ is finite and X is a finite non-empty set of d -tuples — an **index**. Coordinates are written x_i for $1 \leq i \leq d$, and $\pi_i, \pi_{ij}, \pi[F]$ denote projection onto coordinate i , onto the ordered pair (i, j) , and onto a subset F of the coordinates.

D1 (observed alphabet). $A_i := \pi_i(X)$, the set of values X realises at coordinate i . Each A_i carries a total order inherited from its values, so each is a **finite chain**.

D2 (box). $\text{Box}(X) := \prod_i A_i$, the product over $i = 1, \dots, d$. As a product of chains it is a lattice under the coordinate-wise operations

$$(x \wedge y)_i = \min(x_i, y_i), (x \vee y)_i = \max(x_i, y_i),$$

and since any subset of a chain is closed under $\min$ and $\max$ of its own elements, $\text{Box}(X)$ is closed under both.

D3 (generated sublattice, or lattice hull). $\langle X \rangle :=$ the smallest subset of $\text{Box}(X)$ containing X and closed under $\wedge$ and $\vee$. It exists: the intersection of any family of $\wedge, \vee$ -closed subsets is $\wedge, \vee$ -closed, and $\text{Box}(X)$ is one such superset. Because $\wedge$ and $\vee$ act coordinatewise, no coordinate value outside A_i is ever created, so $\langle X \rangle \subseteq \text{Box}(X)$ as stated.

D4 (boundary function). For $i \neq j$ and $a \in A_j$,

$$\varphi_{ij}(a) := \max \{ y_i : y \in X, y_j \leq a \}.$$

When the index must be named, the boundary function of X is written $\varphi_{ij}[X]$.

D5 (staircase). $\mathcal{R}(X) := \{ x \in \text{Box}(X) : x_i \leq \varphi_{ij}(x_j) \text{ for all } i \neq j \}$.

The word is that of Deville, Barette and Van Hentenryck (1999): their connected row-convex constraints are the relations cut by such bounds, and §3 says how their class and this operator meet.

D6 (closed). X is **closed** if $x \wedge y \in X$ and $x \vee y \in X$ for all $x, y \in X$ — that is, if X is a sublattice of $\text{Box}(X)$. Equivalently, of any product of chains containing it, since the operations are coordinatewise.

D7 (closure defect). $E(X) := |\mathcal{R}(X)| - |X|$.

D8 (seed). $G \subseteq X$ is a **seed** of a closed index X if $\mathcal{R}(G) = X$. The **seed number** $\text{seed}(X)$ is the least cardinality of a seed.

By Theorem 2, $\mathcal{R}(G) = \langle G \rangle$, so G is a seed of X exactly when G generates X as a lattice, and $\text{seed}(X)$ is the least number of generators of the lattice X — a classical quantity, for which the paper keeps the name seed because Theorem 13 computes it from the boundary functions.

The ambient regime. Every set is read in **its own box**: $\mathcal{R}(X)$ is cut out of $\text{Box}(X)$, which is determined by X itself. This is the regime in which D1 makes the maxima of D4 attained, and it is the regime of every statement below. The alternative — reading X and a superset in one declared ambient that exceeds either observed box — changes the operator and is not used here.

Notation. $\lfloor \cdot \rfloor$ is the floor. $C(n, k)$ is the binomial coefficient. For a family $\mathcal{F}$ of subsets of a finite set U , $\mathcal{F}$ is read as an index over U by identifying each member with its characteristic vector in $\{0,1\}^U$; $\text{Cl}(U)$ denotes the family of all closed subsets of the box U , the empty set included.

§2 · The defect

The two readings of D1 and D4 are the whole of the reconstruction, and nothing else enters it. $\mathcal{R}(X)$ is what a reader could rebuild from the cells alone.

Lemma 1 (ϕ is total and isotone). For every $i \neq j$ and $a \in A_j$, $\phi_{ij}(a)$ is defined; and $a \leq a'$ in A_j implies $\phi_{ij}(a) \leq \phi_{ij}(a')$.

Proof. By D1 there is $y \in X$ with $y_j = a$, so $\{y \in X : y_j \leq a\}$ is non-empty and its image under π_i is a non-empty finite set of values in the chain A_i , which therefore has a maximum. If $a \leq a'$, the set $\{y \in X : y_j \leq a\}$ is contained in $\{y \in X : y_j \leq a'\}$, so the maximum over the larger set is at least the maximum over the smaller. ■ **PROVED.**

Lemma 2 (witness form). For $x \in \text{Box}(X)$ and $i \neq j$: $x_i \leq \phi_{ij}(x_j)$ if and only if there exists $y \in X$ with $y_j \leq x_j$ and $y_i \geq x_i$.

Proof. If such a y exists then $\phi_{ij}(x_j) \geq y_i \geq x_i$. Conversely, if $x_i \leq \phi_{ij}(x_j)$ then, the maximum in D4 being attained (Lemma 1), some $y \in X$ has $y_j \leq x_j$ and $y_i = \phi_{ij}(x_j) \geq x_i$. ■ **PROVED.**

Lemma 2 is what makes every claim below decidable by a solver: it removes both the maximum and ϕ from the membership test, leaving a first-order formula in the membership variables of X .

Proposition 1 (the defect's range). $0 \leq E(X) \leq |\text{Box}(X)| - |X|$, and both bounds are attained.

Proof. The lower bound is extensivity (Theorem 1(i), whose proof uses nothing from this section). The upper bound is $\mathcal{R}(X) \subseteq \text{Box}(X)$, which is D5. The lower bound is attained by any closed index, for instance the ordered box below. The upper bound is attained by $X = \{(0,1), (1,0)\}$ in a 2×2 box: both boundary functions are constant at 1, every constraint is vacuous, and $\mathcal{R}(X)$ is all four cells, so $E(X) = 2 = 4 - 2$. ■ **PROVED**, the witness **EXHAUSTIVE** (§9, the seed of the full box 2^2).

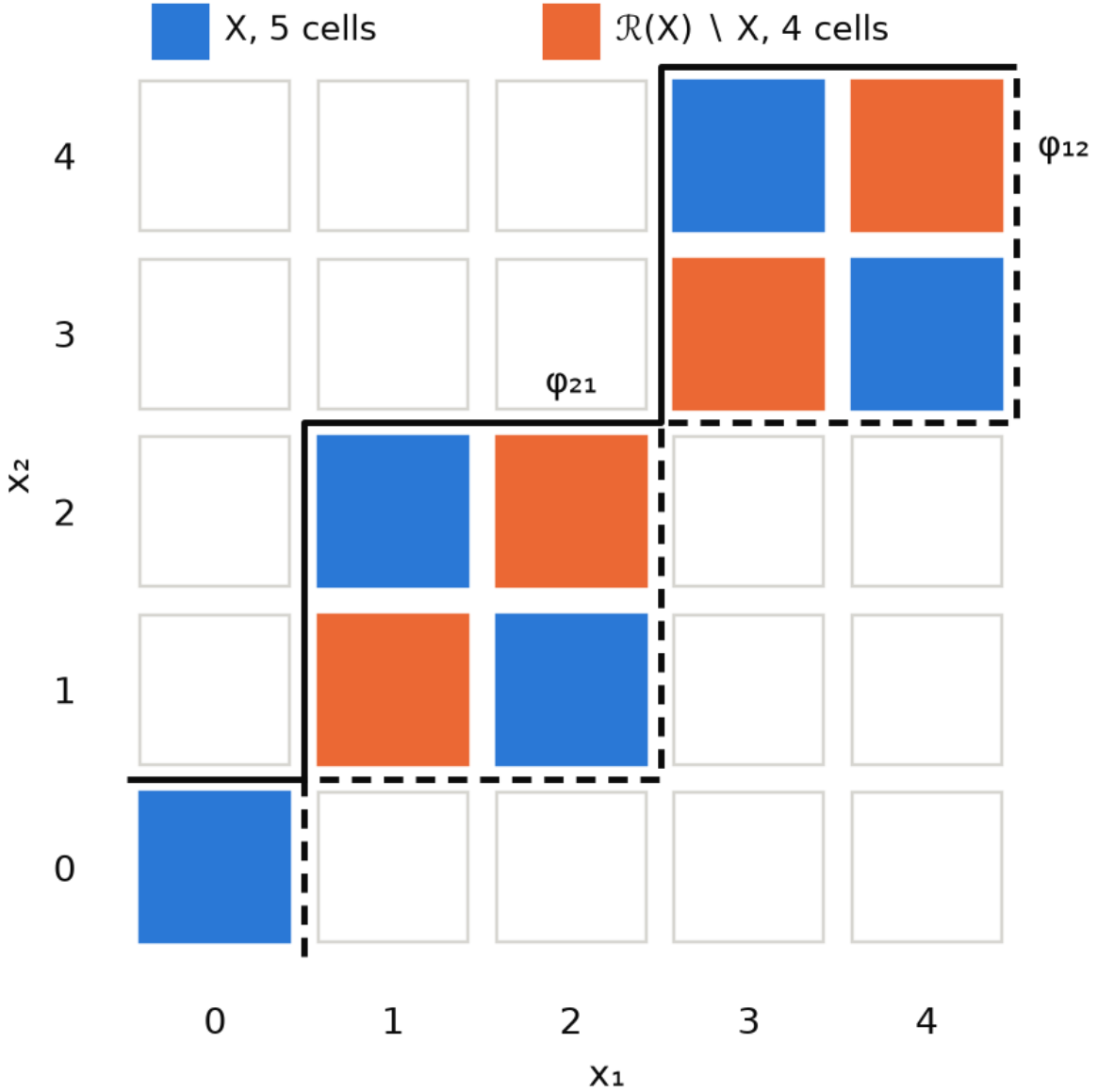


Figure 1. A five-cell index in a 5×5 box (blue) and the four cells its staircase adds (orange): $|\mathcal{R}(X)| = 9$, so $E(X) = 4$. The solid line is φ_{21} , the largest second coordinate reachable at or below a given first coordinate; the dashed line is φ_{12} . $\mathcal{R}(X)$ is everything in the box lying under both, and by Theorem 2 it is the sublattice the five cells generate.

Three indexes measured

The calendar. Take the 365 cells (m, d) of a common year, m the month in its usual order and d the day. Then $|\mathcal{R}(X)| = 372$ and $E(X) = 7$. The seven cells $\mathcal{R}$ admits and the calendar denies are

$$(2, 29) \cdot (2, 30) \cdot (2, 31) \cdot (4, 31) \cdot (6, 31) \cdot (9, 31) \cdot (11, 31)$$

and they are exactly the content of the rhyme taught beside the calendar. The mechanism is a single failure of monotonicity: month length is not monotone in month number, because February is short and stands second, so φ_{21} — the longest day reachable at or before month m , in the coordinate order (month, day) — is already 31 at $m = 1$ and stays there. Relabel the months in increasing order of length and the day-fibre extremes become isotone

— the lower extreme is constantly 1 and the upper is the month length — so by Theorem 5 the relabelled index is closed and E is 0 over the same 365 cells. The relabelled calendar is self-defining and unusable. $E = 7$ is the price of keeping January first.

The periodic table. Take the eighteen-column table with the f block set aside — the 118 elements less the 28 of the two inner series Ce–Lu and Th–Lr, La and Ac standing in group 3 so that periods 6 and 7 are full: 90 cells on the coordinates (period, group), helium in group 18. Then $|\mathcal{R}(X)| = 126$ and $E(X) = 36$. The thirty-six are exactly the gaps in the short periods — period 1 groups 2 to 17, sixteen cells, and periods 2 and 3 groups 3 to 12, twenty cells. They are not a defect in the elements. Period and group locate an element; they do not encode the shell structure that forbids most of these cells and defers the rest — why period 1 holds two elements and period 4 holds eighteen — and that information travels alongside the table rather than in it.

The same 90 cells with helium drawn in group 2 give $E = 20$. The mechanism is visible in one boundary function: $\varphi_{21}(1)$ in the coordinate order (period, group) — the largest group reachable in period 1 —, is 18 under the standard placement and 2 under the alternative, so the whole first row of gaps exists only under the first. Helium’s position — group 18, where IUPAC places it, or group 2, above beryllium — is a standing question of periodic-table design (Scerri 2020; Grochala 2018), and E prices the two answers: the difference is sixteen cells.

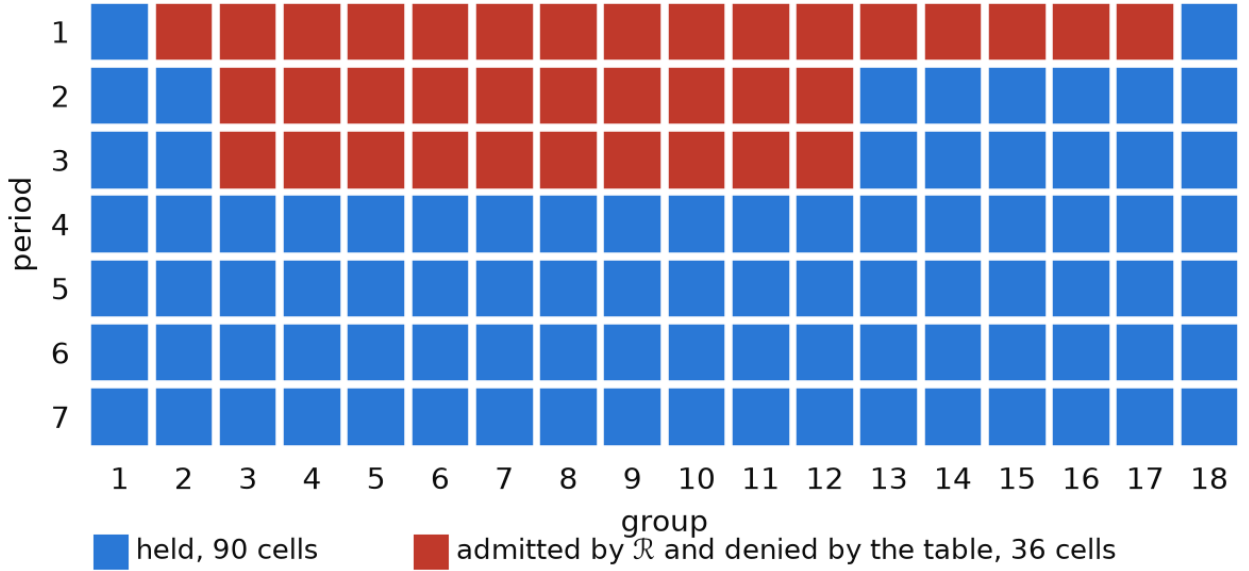


Figure 2. The eighteen-column table on the coordinates (period, group), f block detached. Blue: the 90 cells the table holds. Red: the 36 its own coordinates admit and it denies — period 1 groups 2 to 17, and periods 2 and 3 groups 3 to 12. A reader given only the occupied cells would reconstruct all 126. The figure is computed from the same 90 cells the check closes.

An ordered box. Take the triples (l, w, h) with $6 \geq l \geq w \geq h \geq 1$: 56 cells, $E = 0$. The index is closed, and the reason is visible in its presentation: each of the two inequalities bounds one coordinate by a monotone function of one other, which is exactly the shape $D5$ cuts with, so the recovered bounds reproduce the defining ones and nothing is added. This is the ordered simplex $D(3, 6)$ of §9 up to a shift of values, and Theorem 14 proves its closure for every d and c .

EXHAUSTIVE. All four measurements are recomputed cell by cell: 365 cells and the seven named; 90 cells, $|\mathcal{R}| = 126$, the 36 named as two blocks; the same 90 with helium at group 2, $E = 20$; 56 cells, $E = 0$.

§3 · The closure theorem

Lemma 3 ($\mathcal{R}$ moves neither the box nor the boundary). $\text{Box}(\mathcal{R}(X)) = \text{Box}(X)$, and $\varphi_{ij}[\mathcal{R}(X)] = \varphi_{ij}[X]$ on A_j for every $i \neq j$.

Proof. $X \subseteq \mathcal{R}(X) \subseteq \text{Box}(X)$ — the first inclusion is Theorem 1(i) below, whose proof does not use this lemma, and the second is D5. Applying π_i to the chain of inclusions gives $A_i \subseteq \pi_i(\mathcal{R}(X)) \subseteq A_i$, so the alphabets and hence the boxes agree. For the boundary: fix $i \neq j$ and $a \in A_j$. Since $X \subseteq \mathcal{R}(X)$, every candidate for $\varphi_{ij}[X](a)$ is a candidate for $\varphi_{ij}[\mathcal{R}(X)](a)$, so $\varphi_{ij}[X](a) \leq \varphi_{ij}[\mathcal{R}(X)](a)$. Conversely let $y \in \mathcal{R}(X)$ with $y_j \leq a$. Then $y_i \leq \varphi_{ij}[X](y_j) \leq \varphi_{ij}[X](a)$ by D5 and Lemma 1. So every candidate contributed by $\mathcal{R}(X)$ is bounded by $\varphi_{ij}[X](a)$, and the maximum is too. ■ **PROVED.**

Theorem 1 ($\mathcal{R}$ is a closure operator). For all finite non-empty X, Y :

- (i) *extensive* — $X \subseteq \mathcal{R}(X)$;
- (ii) *monotone* — $X \subseteq Y$ implies $\mathcal{R}(X) \subseteq \mathcal{R}(Y)$;
- (iii) *idempotent* — $\mathcal{R}(\mathcal{R}(X)) = \mathcal{R}(X)$.

Proof. (i) Let $x \in X$. Then $x \in \text{Box}(X)$ by D1, and for any $i \neq j$ the cell x is its own witness: $x_j \leq x_j$ and $x_i \geq x_i$, so $\varphi_{ij}(x_j) \geq x_i$ by Lemma 2.

(ii) Let $X \subseteq Y$ and $x \in \mathcal{R}(X)$. First $x \in \text{Box}(X) \subseteq \text{Box}(Y)$, since $\pi_i(X) \subseteq \pi_i(Y)$. Next fix $i \neq j$. By Lemma 2 there is $y \in X$ with $y_j \leq x_j$ and $y_i \geq x_i$; that same y lies in Y , so $x_i \leq \varphi_{ij}[Y](x_j)$ by Lemma 2 again. Hence $x \in \mathcal{R}(Y)$.

(iii) By Lemma 3, $\mathcal{R}(X)$ has the same box and the same boundary functions as X . D5 cuts $\mathcal{R}(\mathcal{R}(X))$ out of $\text{Box}(\mathcal{R}(X)) = \text{Box}(X)$ by the conditions $x_i \leq \varphi_{ij}[\mathcal{R}(X)](x_j) = \varphi_{ij}[X](x_j)$, which are the conditions defining $\mathcal{R}(X)$. The two sets are cut out of the same box by the same inequalities, so they are equal. ■ **PROVED**, and **MACHINE-CHECKED** over every subset of the boxes 3×3 , 4×4 , $2 \times 2 \times 2$ and $3 \times 3 \times 3$, and **EXHAUSTIVE** over 74,569 subsets of seven boxes; (ii) additionally over all 26,206 nested pairs of four boxes.

Lemma 4 ($\mathcal{R}(X)$ is a sublattice). $\mathcal{R}(X)$ is closed under $\vee$ and $\wedge$.

Proof. Let $x, z \in \mathcal{R}(X)$. Both $x \vee z$ and $x \wedge z$ lie in $\text{Box}(X)$, since each coordinate of either is one of x_i, z_i , both in A_i . Fix $i \neq j$.

Join. Write $u = x \vee z$. Then $u_i = \max(x_i, z_i)$, which is x_i or z_i ; say $u_i = x_i$. Also $u_j = \max(x_j, z_j) \geq x_j$. By Lemma 1, $\varphi_{ij}(u_j) \geq \varphi_{ij}(x_j) \geq x_i = u_i$.

Meet. Write $v = x \wedge z$. Then $v_j = \min(x_j, z_j)$, which is x_j or z_j ; say $v_j = x_j$. Then $\varphi_{ij}(v_j) = \varphi_{ij}(x_j) \geq x_i \geq \min(x_i, z_i) = v_i$.

Both cases hold for every ordered pair, so $u, v \in \mathcal{R}(X)$. ■ **PROVED**, and **MACHINE-CHECKED** over every subset of 3×3 , 4×4 , $2 \times 2 \times 2$, $3 \times 3 \times 3$.

The converse direction — that $\mathcal{R}(X)$ is not merely a sublattice but the *smallest* one — is the substance. At $d = 2$ it has a short constructive proof.

Lemma 5 (the four-witness construction, $d = 2$). Let $d = 2$ and $(a, b) \in \mathcal{R}(X)$. Then there are four cells of X from which (a, b) is built by two joins and one meet.

Proof. Since $a \in A_1$ there is $y \in X$ with $y_1 = a$; since $b \in A_2$ there is $z \in X$ with $z_2 = b$. Since $b \leq \varphi_{21}(a)$ and the maximum is attained, there is $u \in X$ with $u_1 \leq a$ and $u_2 = \varphi_{21}(a) \geq b$. Since $a \leq \varphi_{12}(b)$ there is $w \in X$ with $w_2 \leq b$ and $w_1 = \varphi_{12}(b) \geq a$. Put

$$\alpha := y \vee u, \gamma := z \vee w.$$

Then $\alpha_1 = \max(a, u_1) = a$ because $u_1 \leq a$, and $\alpha_2 = \max(y_2, u_2) \geq u_2 \geq b$. Symmetrically $\gamma_2 = \max(b, w_2) = b$ and $\gamma_1 = \max(z_1, w_1) \geq w_1 \geq a$. Hence

$$\alpha \wedge \gamma = (\min(a, \gamma_1), \min(\alpha_2, b)) = (a, b).$$

All four of y, u, z, w lie in X . ■ **PROVED**, and **EXHAUSTIVE**: the construction is built and verified on every one of 815,072 cells of $\mathcal{R}(X)$, over every subset X of a 3×3 and a 4×4 box, with 0 failures.

Theorem 2 ($\mathcal{R}(X)$ is the hull). Let S be a sublattice of a finite product of chains with $X \subseteq S$. Then $\mathcal{R}(X) \subseteq S$. Consequently $\mathcal{R}(X) = \langle X \rangle$.

Proof. It suffices to prove $\mathcal{R}(X) \subseteq \langle X \rangle$. For if S is a sublattice of a product of chains with $X \subseteq S$, then $S \cap \text{Box}(X)$ is a sublattice of $\text{Box}(X)$ containing X — an intersection of two sublattices, $\text{Box}(X)$ being one — so $\langle X \rangle \subseteq S \cap \text{Box}(X) \subseteq S$ by D3. Write $T := \langle X \rangle$.

Case $d = 2$. Let $x \in \mathcal{R}(X)$. Lemma 5 exhibits $y, u, z, w \in X$ with $x = (y \vee u) \wedge (z \vee w)$. All four lie in T , and T is closed under $\vee$ and $\wedge$, so $x \in T$.

Case $d \geq 3$. Fix $x \in \mathcal{R}(X)$, and let $i < j$ be any pair of coordinates.

Step 1: $\pi_{ij}(x) \in \mathcal{R}(\pi_{ij} X)$. The set $\pi_{ij}(X)$ is an index on two coordinates, and its own data are inherited: its observed alphabets are A_i and A_j , and its boundary functions are $\phi_{ij}[X]$ and $\phi[X]_{ji}$, because the condition $y_j \leq a$ and the value y_i both depend on y only through the two coordinates kept. Now $x \in \text{Box}(X)$ gives $\pi_{ij}(x) \in A_i \times A_j = \text{Box}(\pi_{ij} X)$, and $x \in \mathcal{R}(X)$ gives $x_i \leq \phi_{ij}[X](x_j)$ and $x_j \leq \phi[X]_{ji}(x_i)$, which are the two conditions of D5 in dimension two. So $\pi_{ij}(x) \in \mathcal{R}(\pi_{ij} X)$.

Step 2: $\mathcal{R}(\pi_{ij} X) \subseteq \pi_{ij}(T)$. The image $\pi_{ij}(T)$ is a sublattice of $A_i \times A_j$: for $p, q \in T$, $\pi_{ij}(p) \vee \pi_{ij}(q) = \pi_{ij}(p \vee q) \in \pi_{ij}(T)$ because the join is coordinatewise, and likewise for the meet. (§6, Theorem 8, states this for an arbitrary set of coordinates.) It contains $\pi_{ij}(X)$. So by the case $d = 2$, applied to the index $\pi_{ij}(X)$ and the sublattice $\pi_{ij}(T)$, we get $\mathcal{R}(\pi_{ij} X) \subseteq \pi_{ij}(T)$. With Step 1, $\pi_{ij}(x) \in \pi_{ij}(T)$ for every pair $i < j$.

Step 3: the lift. T is a sublattice of the finite product $\prod_i A_i$ of lattices. The lattice median $m(p, q, r) = (p \wedge q) \vee (q \wedge r) \vee (r \wedge p)$ satisfies $m(p, p, q) = m(p, q, p) = m(q, p, p) = p$, so it is a majority term for the variety of lattices. By **Baker and Pixley's interpolation theorem (1975)** — CITED — a subalgebra of a finite product of algebras in a variety with a majority term is **2-decomposable**: it equals the set of tuples all of whose two-fold projections are two-fold projections of its members. Hence

$$T = \{ p \in \prod_i A_i : \pi_{ij}(p) \in \pi_{ij}(T) \text{ for all } i < j \},$$

and Step 2 places x in that set. So $x \in T$.

In both cases $\mathcal{R}(X) \subseteq T = \langle X \rangle$; with Lemma 4 and Theorem 1(i), $\mathcal{R}(X)$ is a sublattice containing X , so $\langle X \rangle \subseteq \mathcal{R}(X)$ and the two are equal. ■ **PROVED at $d = 2$; PROVED with one CITED step at $d \geq 3$; MACHINE-CHECKED** over every subset of 3×3 , 4×4 , $2 \times 2 \times 2$, $3 \times 3 \times 3$ and $2 \times 2 \times 2 \times 2$, the sublattice S quantified over as well; **EXHAUSTIVE** $\mathcal{R}(X) = \langle X \rangle$ on every non-empty subset of every box of at most twelve cells among 2×2 , 3×2 , $2 \times 2 \times 2$, 3×3 , 4×3 , $2 \times 2 \times 3$, $2 \times 2 \times 2 \times 2$.

Theorem 3 (the characterisation). X is closed if and only if $X = \mathcal{R}(X)$ — equivalently, if and only if $E(X) = 0$.

Proof. If $X = \mathcal{R}(X)$ then X is a sublattice by Lemma 4. Conversely if X is closed then X is a sublattice containing X , so $\mathcal{R}(X) \subseteq X$ by Theorem 2; with Theorem 1(i), $X = \mathcal{R}(X)$. The restatement in terms of E is D7 with Theorem 1(i), which gives $E(X) \geq 0$ with equality exactly at $X = \mathcal{R}(X)$. ■ **PROVED, MACHINE-CHECKED** over every subset of 3×3 , 4×4 , $2 \times 2 \times 2$, $3 \times 3 \times 3$, and **EXHAUSTIVE** over 74,569 subsets of seven boxes.

What is new here and what is not

The operator $\mathcal{R}$ is a closure operator in the sense of **Moore (1910)**, and the fixed sets of any closure operator form a Moore family — **Ward (1942)**, surveyed in **Caspard and Monjardet (2003)**. Nothing about $\mathcal{R}$ in particular is needed for §5's first clause. Theorem 2's content above $d = 2$ is that a sublattice of a finite product of lattices is determined by its two-fold projections: the majority-term case of **Baker and Pixley's (1975)** interpolation theorem. **Bergman (1977)** named the property — d -fold projection determination — and proved its converse, that it characterises the varieties with a $(d+1)$ -ary near-unanimity term; for sublattices of a product of chains the forward direction is also in **Topkis (1976)**, in **Veinott (1989)**, and in **Queyranne and Tardella (2008)**, who state it for the sublattice hull in a product directly. In the language of constraint satisfaction, $E(X) = 0$ says that X is the solution set of a binary network whose constraints are the recovered staircases — in the terms of **Montanari (1974)**, X is binary-decomposable into (α, β) -monotone constraints with $\alpha, \beta \in \{\leq, \geq\}$ — and that network, path consistent and row convex, is then globally consistent by **van Beek and Dechter (1995)**. Global consistency is a property of a network and not of a relation, and it is not what $E = 0$ names: the network of recovered staircases is globally consistent for every X — for $X = \{(0,1), (1,0)\}$ it is the full 2×2 box, globally consistent, at $E = 2$. What $E = 0$ names is the decomposition. The class of constraints ϕ recovers is the **staircase**, or connected row-convex, class of **Déville, Barette and Van Hentenryck (1999)**.

Freuder (1982) is not one of these. That paper gives a *sufficient condition for backtrack-free search*: a constraint network whose graph has width w admits backtrack-free search if it is strongly $(w+1)$ -consistent, and it says explicitly that even width 2 demands strong 3-consistency. That is a statement about the cost of finding a solution, not about a network's solution set being determined by its binary projections. The decomposition is Montanari's, and the consistency certificate for the row-convex class is van Beek and Dechter's.

What this paper contributes is the reading of $|\mathcal{R}(X)| - |X|$ as a *defect of an index* — a measurable price of a layout — together with the consequences of §4 to §9. Among those, §9 is placed exactly: Theorem 13 (the seed as a set cover) and Theorem 14 (the simplex law) are the paper's; Theorem 15 (the full-box law) is Czédli's (2023a; 2023b), and it is marked CITED where it is stated.

§4 • The staircase algebra

A closed index is not merely a sublattice; it is the solution set of a canonical system of inequalities, and the system can be read back off the cells.

Theorem 4 (the boundary functions are the least representation). Let $\psi = (\psi_{ij})$, $i \neq j$, be any family of isotone functions $\psi_{ij} : A_j \rightarrow \mathbb{Z}$ such that

$$X = \{ x \in \text{Box}(X) : x_i \leq \psi_{ij}(x_j) \text{ for all } i \neq j \}.$$

Then $\phi_{ij}(a) \leq \psi_{ij}(a)$ for every $i \neq j$ and $a \in A_j$.

Proof. Fix $i \neq j$ and $a \in A_j$. By Lemma 1 the maximum in D4 is attained: there is $y \in X$ with $y_j \leq a$ and $y_i = \phi_{ij}(a)$. Since $y \in X$, the hypothesis gives $y_i \leq \psi_{ij}(y_j)$, and ψ_{ij} is isotone with $y_j \leq a$, so $\psi_{ij}(y_j) \leq \psi_{ij}(a)$. Chaining, $\phi_{ij}(a) = y_i \leq \psi_{ij}(a)$. ■ **PROVED**, and **MACHINE-CHECKED** over every subset of 3×3 and $2 \times 2 \times 2$, with ψ quantified over all integer-valued isotone systems.

So among all isotone bound systems that cut X out of its box, ϕ is the pointwise smallest. Two readings follow. First, the **pointwise-minimum normal form**: D5 may be written

$$\mathcal{R}(X) = \{ x \in \text{Box}(X) : x_i \leq \min \{ \phi_{ij}(x_j) : j \neq i \} \text{ for every } i \},$$

which is D5 rearranged, and the minimum is genuinely pointwise: which of the $d - 1$ bounds on a coordinate is the binding one may differ from cell to cell, so the recovered system carries a bound for every ordered pair whether or not the presentation did. Second, the map from a presentation to its normal form runs one way only: $\mathcal{R}$ recovers the bounds but not the order in which they were imposed, so many presentations share a normal form.

Theorem 5 (closed sets at $d = 2$ are bands). Let $d = 2$, let S be a finite non-empty set of pairs with observed alphabets $A_1 = \pi_1(S)$ and $A_2 = \pi_2(S)$, and for $s \in A_1$ write $F(s) = \{ t : (s, t) \in S \}$. Then S is closed if and only if there are isotone functions $L, U : A_1 \rightarrow A_2$ with $L \leq U$ such that

$$S = \{ (s, t) \in A_1 \times A_2 : L(s) \leq t \leq U(s) \};$$

and in that case $L(s) = \min F(s)$ and $U(s) = \max F(s)$.

Proof. Necessity. Let S be closed, so every $F(s)$ for $s \in A_1$ is non-empty; put $L(s) = \min F(s)$, $U(s) = \max F(s)$.

Fix $s \in A_1$ and take $t \in A_2$ with $L(s) \leq t \leq U(s)$; we show $(s, t) \in S$. Since $t \in A_2$ there is $s' \in A_1$ with $(s', t) \in S$. If $s' \leq s$, then $(s, L(s)) \vee (s', t) = (s, \max(L(s), t)) = (s, t)$, which lies in S by closure. If $s' > s$, then $(s, U(s)) \wedge (s', t) = (s, \min(U(s), t)) = (s, t)$, again in S . So S contains the whole band, and it is contained in it by the definitions of L and U .

Isotonicity: take $s < s'$ in A_1 . Then $(s, U(s)) \vee (s', L(s')) = (s', \max(U(s), L(s')))$ lies in S , so $\max(U(s), L(s')) \leq U(s')$, and since the maximum is at least $U(s)$ this gives $U(s) \leq U(s')$. For L , split on the comparison. If $U(s) \leq L(s')$ then $L(s) \leq U(s) \leq L(s')$. Otherwise $\min(U(s), L(s')) = L(s')$, and $(s, U(s)) \wedge (s', L(s')) = (s, L(s'))$ lies in S , so $L(s') \geq L(s)$ by the definition of $L(s)$. Either way $L(s) \leq L(s')$.

Sufficiency. Let $L \leq U$ be isotone and W the band they define. Take $(s, t), (s', t') \in W$ with $s \leq s'$. The join is $(s', \max(t, t'))$. Lower: $L(s') \leq t' \leq \max(t, t')$. Upper: $t \leq U(s) \leq U(s')$ and $t' \leq U(s')$, so $\max(t, t') \leq U(s')$. The meet is $(s, \min(t, t'))$. Lower: $L(s) \leq t$ and $L(s) \leq L(s') \leq t'$, so $L(s) \leq \min(t, t')$. Upper: $\min(t, t') \leq t \leq U(s)$. Both lie in W . ■
PROVED, MACHINE-CHECKED in both directions over every subset of 3×3 , 4×4 and 3×5 , and **EXHAUSTIVE** on all 3,713 closed subsets of a 4×4 and a 3×5 box, with 0 failures.

The algebra, written out

Theorem 5 says that at $d = 2$ a closed index is a system of two inequalities, and the inequalities can be printed. A worked instance, nine cells on coordinates (s, t) :

$$\lfloor s/2 \rfloor \leq t \leq s + \lfloor s/3 \rfloor, s \in \{0, 1, 2, 3\}.$$

The cells are $(0,0), (1,0), (1,1), (2,1), (2,2), (3,1), (3,2), (3,3), (3,4)$ — nine, with $E = 0$. Reading the fibre extremes back off those nine cells returns $L = (0, 0, 1, 1)$ and $U = (0, 1, 2, 4)$, which are $\lfloor s/2 \rfloor$ and $s + \lfloor s/3 \rfloor$ at $s = 0, 1, 2, 3$; and the boundary function φ_{21} recovered by D4 in the coordinate order (s, t) equals U at every s . The two floor expressions are not asserted to exist and then left: they are what the reconstruction returns.

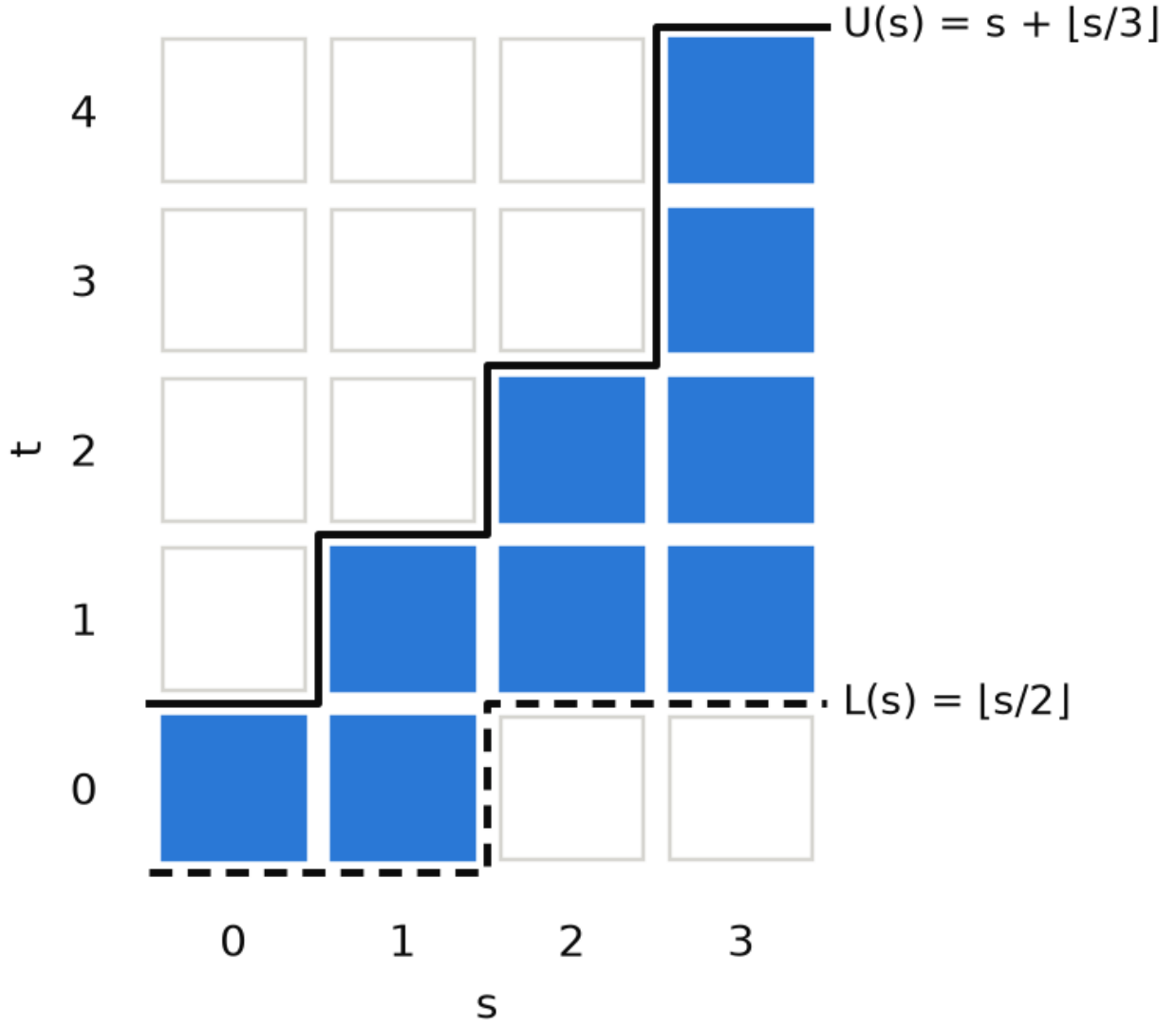


Figure 3. The nine-cell staircase between $L(s) = \lfloor s/2 \rfloor$ (dashed) and $U(s) = s + \lfloor s/3 \rfloor$ (solid). The grid is the box $A_1 \times A_2 = \{0, \dots, 3\} \times \{0, \dots, 4\}$ of the nine cells. Both bounds are isotone and $L \leq U$, so by Theorem 5 the band is closed; $E = 0$, and the fibre extremes read off the nine cells are $L = (0, 0, 1, 1)$ and $U = (0, 1, 2, 4)$.

§5 • The family of closed sets

Fix a finite box U — a product of chains, read as a set of cells — and let $\text{Cl}(U)$ be the family of all closed subsets of U , the empty set included, and write $n = |U|$ for its number of cells.

Theorem 6 (Cl(U) is a Moore family). $U \in \text{Cl}(U)$, and $S, T \in \text{Cl}(U)$ implies $S \cap T \in \text{Cl}(U)$. The family is **not** closed under union.

Proof. U is a product of chains and so closed under the coordinatewise operations (D2). If S and T are closed and $x, y \in S \cap T$, then $x \wedge y \in S$ and $x \wedge y \in T$, so $x \wedge y \in S \cap T$; likewise for $\vee$. The failure of union-closure is exhibited below. ■ **PROVED**, and **MACHINE-CHECKED** — as the statement that the fixed points of $\mathcal{R}$ are closed under intersection — over every subset of 3×3 , 4×4 , $2 \times 2 \times 2$ and $3 \times 3 \times 3$.

REFUTATION (union). In the 2×2 box, $S_1 = \{(0,0), (1,0)\}$ and $S_2 = \{(0,0), (0,1)\}$ are both closed — each is a two-element chain — but $S_1 \cup S_2$ omits $(1,0) \vee (0,1) = (1,1)$ and is not closed. Their intersection $\{(0,0)\}$ is closed, as Theorem 6 requires. **REFUTATION**, re-verified.

That $\text{Cl}(U)$ is a Moore family is the standard fact about any closure operator (Moore 1910; Ward 1942), and it is worth stating here because the asymmetry is the operator's own, one level up: $\mathcal{R}$ carries meets and refuses joins, and its family of fixed points does the same. Measured exhaustively:

ambient U	cells	$ \text{Cl}(U) $	non-empty	pairs closed under $\cap$	pairs closed under $\cup$	meet-irreducible	$E(\text{Cl}(U))$
2×2	4	13	12	78 / 78	71 / 78 (91.0%)	6	3
3×2	6	38	37	703 / 703	577 / 703 (82.1%)	9	26
$2 \times 2 \times 2$	8	74	73	2,701 / 2,701	1,765 / 2,701 (65.3%)	12	182
3×3	9	147	146	10,731 / 10,731	7,380 / 10,731 (68.8%)	14	365
4×3	12	506	505	127,765 / 127,765	74,266 / 127,765 (58.1%)	19	3,590
$2 \times 2 \times 3$	12	320	319	51,040 / 51,040	25,147 / 51,040 (49.3%)	17	3,776
$2 \times 2 \times 2 \times 2$	16	732	731	267,546 / 267,546	87,612 / 267,546 (32.7%)	20	64,804

Table 1. Every unordered pair of distinct members of $\text{Cl}(U)$ tested under intersection and under union. Intersection never escapes. Union does, on the complement of the fraction shown — from 9.0% of pairs at the smallest ambient to 67.3% at the largest — and the escape rate grows with the ambient. Restricted to the non-empty members the union rates are 89.4%, 81.1%, 64.4%, 68.3%, 58.0%, 49.0% and 32.6%. Meet-irreducible counts the members other than U that are not the intersection of the members strictly above them; a Moore family is generated under intersection by those members together with U , the empty intersection. The counts are reported and no law is claimed for them.

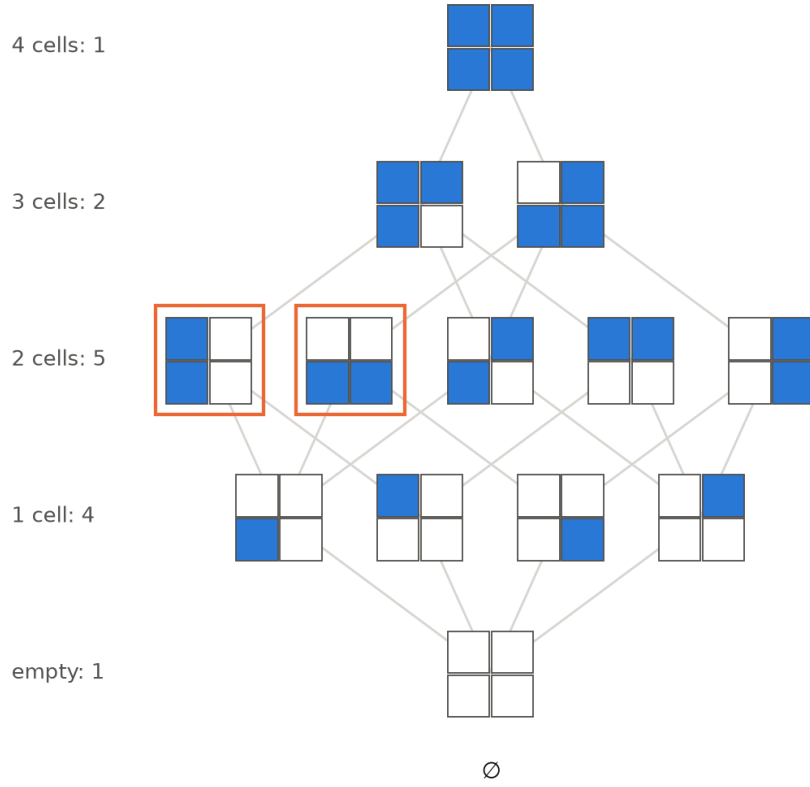


Figure 4. The thirteen closed subsets of the 2×2 box, ordered by inclusion; each member is drawn as the box with its cells filled. Twelve are non-empty. The two ringed members are the witness of the union refutation: both are closed, their union is not, their intersection is.

Proposition 2 (the closed sets separate the cells). For any two distinct cells $p \neq q$ of U there is a member of $\text{Cl}(U)$ containing p and not q .

Proof. $\{p\}$ is closed: $p \wedge p = p \vee p = p$. **■ PROVED**, and **EXHAUSTIVE** at all seven ambients of Table 1, where separation is verified over every ordered pair of cells.

Theorem 7 (the family's own defect is exact). Read $\text{Cl}(U)$ as an index over the coordinate set U , each member identified with its characteristic vector in $\{0,1\}^U$. Then $\mathcal{R}(\text{Cl}(U)) = \{0,1\}^U$, the whole power set, and therefore

$$E(\text{Cl}(U)) = 2^n - |\text{Cl}(U)|, \quad n = |U|.$$

Proof. Alphabets. Fix a cell $p \in U$. The empty set has 0 at coordinate p and U has 1, and both are members (Theorem 6), so the observed alphabet at every coordinate is $\{0, 1\}$ and $\text{Box}(\text{Cl}(U)) = \{0,1\}^U$.

Boundaries. Fix $p \neq r$ in U . Since the alphabet is $\{0,1\}$, the only non-vacuous value of the boundary function is at 0:

$$\varphi_{pr}(0) = \max \{ S_p : S \in \text{Cl}(U), S_r \leq 0 \} = \max \{ S_p : S \in \text{Cl}(U), r \notin S \}.$$

By Proposition 2 there is a closed S with $p \in S$ and $r \notin S$, so this maximum is 1. And $\varphi_{pr}(1) = 1$ likewise, φ being isotone and bounded by 1. So every boundary function is constant at 1, every constraint $x_p \leq \varphi_{pr}(x_r)$ reads $x_p \leq 1$ and is vacuous, and $\mathcal{R}(\text{Cl}(U))$ is the whole box $\{0,1\}^U$. The defect is then $2^n - |\text{Cl}(U)|$ by D7. **■ PROVED**, and **EXHAUSTIVE** at all seven ambients of Table 1, where $|\mathcal{R}(\text{Cl}(U))|$ is computed directly, by closing the family as an index over its 2^n candidate cells — 65,536 of them at $2 \times 2 \times 2 \times 2$ — and found equal to 2^n in every case, so the column $E(\text{Cl}(U))$ is a measurement and not the identity restated.

Theorem 7 sharpens the previous paragraph into a statement with no slack in it. Every member of $\text{Cl}(U)$ has $E = 0$, by Theorem 3. The family itself has the largest defect its ambient permits: an operator all of whose bounds are trivial admits everything, so the family's closure is not merely bigger than the family, it is the entire power set. At sixteen cells that is 64,804 admitted cells against 732 members. So the family of closed indexes is itself an open index, and maximally so — and it is the family every closed index belongs to. Nothing in the proof uses $\mathcal{R}$ beyond Proposition 2: any family of subsets of U that contains $\emptyset$ and U and separates the points of U has every pairwise bound constant at 1, and its closure is the whole power set. Theorem 7 records that $\text{Cl}(U)$ is such a family, and the count is what the identity then reads off.

§6 • Projections

Theorem 8 (projections of closed sets are closed). Let X be closed and F a non-empty set of coordinates. Then $\pi[F](X)$ is closed.

Proof. Take $x, y \in \pi[F](X)$ and lift them to $x', y' \in X$. Since X is closed, $x' \vee y' \in X$. The join is coordinatewise and projection merely drops coordinates, so

$$\pi[F](x' \vee y') = \pi[F](x') \vee \pi[F](y') = x \vee y,$$

which therefore lies in $\pi[F](X)$. The same argument with $\wedge$ throughout gives the meet. ■ **PROVED, MACHINE-CHECKED** for the projection of every fixed point onto its first two coordinates and onto its first and third, over the boxes $2 \times 2 \times 2$, $3 \times 3 \times 3$ and $2 \times 3 \times 4$, and **EXHAUSTIVE** over all 438 projections of every closed subset of a $2 \times 2 \times 2$ box onto every proper set of coordinates.

The converse fails, and it fails often.

REFUTATION (closure is not locally determined). In $\{0,1\}^3$ take

$$S = \{ (0,0,0), (0,0,1), (0,1,0), (0,1,1), (1,0,1), (1,1,0) \}.$$

All three two-fold projections of S are closed — each is a closed subset of a 2×2 box — and so are all three one-fold projections, every subset of a chain being closed. Yet $(1,0,1) \vee (1,1,0) = (1,1,1) \notin S$, so S is not closed. **REFUTATION**, re-verified.

The mechanism is visible in the witness: the two offending cells agree on coordinate 1 and disagree on *both* others, so their join disagrees with each of them in a **combination**, and no projection that omits one of the two coordinates can see a combination it does not contain.

The witness is not isolated. Of the subsets of $\{0,1\}^3$ with at least two cells whose three two-fold projections are all closed, there are **117**, and **52 of them — 44.4% — are not closed**. **EXHAUSTIVE** over that family.

Closure is a d -dimensional condition. No projection of it serves as a criterion — not per coordinate, not per pair. This is the exact point at which Theorem 2's route through two-fold projections is *not* reversible as a membership test for an arbitrary set: the two-fold projections determine a **sublattice** (Baker and Pixley's theorem, applied in Theorem 2 to $T = \langle X \rangle$, which is a sublattice by construction), and they do not determine whether an arbitrary set *is* one.

§7 · Adjunction and products

Adjunction never repairs closure

A natural hope for an open index is that adding a derived coordinate will close it. It cannot, and the reason is exact.

Theorem 9 (the graph criterion). Let S be a set of cells in a box and h a map from the box to a chain H . Write $\Gamma(h) = \{ (x, h(x)) : x \in S \}$ for the graph of h over S . Then $\Gamma(h)$ is closed if and only if both

- (a) S is closed, and
- (b) $h(x \vee y) = \max(h(x), h(y))$ and $h(x \wedge y) = \min(h(x), h(y))$ for all $x, y \in S$

— that is, if and only if S is closed and h is a lattice homomorphism on S .

Proof. The operations on the extended box are coordinatewise, so for $x, y \in S$,

$$(x, h(x)) \vee (y, h(y)) = (x \vee y, \max(h(x), h(y))), (x, h(x)) \wedge (y, h(y)) = (x \wedge y, \min(h(x), h(y))).$$

Sufficiency. Assume (a) and (b). Then $x \vee y \in S$ and $\max(h(x), h(y)) = h(x \vee y)$, so the join displayed above is $(x \vee y, h(x \vee y)) \in \Gamma(h)$. The meet likewise.

Necessity. Assume $\Gamma(h)$ is closed. For $x, y \in S$, the join displayed above lies in $\Gamma(h)$, so it equals $(z, h(z))$ for some $z \in S$; comparing the first block of coordinates gives $z = x \vee y$, so $x \vee y \in S$, and comparing the last gives $h(x \vee y) = \max(h(x), h(y))$. The meet gives $x \wedge y \in S$ and $h(x \wedge y) = \min(h(x), h(y))$. ■ **PROVED**, and **MACHINE-CHECKED** as a biconditional over every subset of 3×3 and $2 \times 2 \times 2$, with h quantified over all integer-valued maps into a three-element chain.

Corollary 1 (adjunction never repairs). If $\Gamma(h)$ is closed then S is closed. Contrapositively, an open index is not made closed by adjoining any function of its own coordinates.

Proof. Clause (a) of Theorem 9. ■ **PROVED**.

So the repair routes for an open index are the ones that move cells or relabel them — enlarge, restrict, reorder — and a derived coordinate is not a fourth. The calendar of §2 is repaired by reordering, at the cost of usability.

Projections, minima, maxima and constants are lattice homomorphisms and so are admissible by Theorem 9. Sums, products and differences are not. The difference is the case worth stating exactly, because it is the one a reader is most likely to want.

Proposition 3 (a coordinate difference is inadmissible). Let $h(x) = x_1 - x_2$, and compare cells by the two coordinates h reads: $x \sqsubseteq y$ iff $x_1 \leq y_1$ and $x_2 \leq y_2$ — a partial order when $d = 2$ and, for $d > 2$, a preorder in which cells agreeing on those two coordinates are mutually comparable. Then h is a lattice homomorphism on a set S if and only if S is a chain in $\sqsubseteq$ (any two of its cells comparable) and h is $\sqsubseteq$ -isotone on S .

Proof. *Sufficiency.* Let S be a $\sqsubseteq$ -chain with h isotone, and take $x, y \in S$, say $x \sqsubseteq y$. Then $(x \vee y)_1 = y_1$ and $(x \vee y)_2 = y_2$, so $h(x \vee y) = h(y) = \max(h(x), h(y))$ by isotonicity; and $(x \wedge y)_1 = x_1$, $(x \wedge y)_2 = x_2$, so $h(x \wedge y) = h(x) = \min(h(x), h(y))$.

Necessity. Suppose $x, y \in S$ are $\sqsubseteq$ -incomparable; without loss of generality $x_1 > y_1$ and $x_2 < y_2$. Then

$$h(x \vee y) = \max(x_1, y_1) - \max(x_2, y_2) = x_1 - y_2 < x_1 - x_2 = h(x),$$

the inequality because $y_2 > x_2$. Hence $h(x \vee y) < h(x) \leq \max(h(x), h(y))$, so h is not a join-homomorphism on S . Therefore S is a $\sqsubseteq$ -chain; and then for $x \sqsubseteq y$ the join identity reads $h(y) = \max(h(x), h(y))$, which is $h(x) \leq h(y)$. ■ **PROVED**, and **MACHINE-CHECKED** as a biconditional over every subset of 3×3 , 4×4 and $3 \times 3 \times 3$.

A difference is therefore admissible as a coordinate only on a chain. It fails on anything wider because at a join the two terms take their maxima independently, and the difference of the maxima need not be the maximum of the differences. What survives is weaker and still useful.

Remark (a difference is an interval map). For $h(x) = x_1 - x_2$ and any cells a, b ,

$$\min(h(a), h(b)) \leq h(a \vee b) \leq \max(h(a), h(b)), \text{ and the same for } a \wedge b.$$

Proof. $h(a \vee b) = \max(a_1, b_1) - \max(a_2, b_2)$; take $a_1 \geq b_1$. If $a_2 \geq b_2$ then $h(a \vee b) = a_1 - a_2 = h(a)$. Otherwise $h(a \vee b) = a_1 - b_2$, and $a_1 - b_2 \leq a_1 - a_2 = h(a)$ while $a_1 - b_2 \geq b_1 - b_2 = h(b)$. For the meet, $h(a \wedge b) = \min(a_1, b_1) - \min(a_2, b_2)$; take $a_1 \leq b_1$. If $a_2 \leq b_2$ then $h(a \wedge b) = h(a)$. Otherwise $h(a \wedge b) = a_1 - b_2$, and $a_1 - b_2 \geq a_1 - a_2 = h(a)$ while $a_1 - b_2 \leq b_1 - b_2 = h(b)$. In every case the value lies between $h(a)$ and $h(b)$. ■ **PROVED.**

The image of a join under a difference is not determined; it is bracketed. That is strictly weaker than a homomorphism and strictly stronger than nothing, and it is what closure leaves of a quantity an index cannot carry as a coordinate.

The product rule

Proposition 4 (products factorise). Let P be a non-empty index on a coordinate set I and Q a non-empty index on a disjoint coordinate set J , and let $P \times Q$ denote the index on $I \cup J$ whose cells are the concatenations. Then

$$\mathcal{R}(P \times Q) = \mathcal{R}(P) \times \mathcal{R}(Q), \text{ and } E(P \times Q) = |P| \cdot E(Q) + |Q| \cdot E(P) + E(P) \cdot E(Q).$$

Proof. *Alphabets.* For $i \in I$, $\pi_i(P \times Q) = \pi_i(P)$ because Q is non-empty; likewise on J . So $\text{Box}(P \times Q) = \text{Box}(P) \times \text{Box}(Q)$.

Boundaries. For $i, j \in I$ with $i \neq j$ and $a \in A_j$, the cells of $P \times Q$ whose j -coordinate is at most a are exactly the concatenations of a $y \in P$ with $y_j \leq a$ and a $z \in Q$, and their i -coordinate is y_i . So the boundary function of the product on I is that of P , and likewise on J . For $i \in I$ and $j \in J$, the condition $y_j \leq a$ places no restriction on the I -part, so the boundary function is $\max \pi_i(P)$ at every a , and the constraint $x_i \leq \max \pi_i(P)$ holds on all of $\text{Box}(P \times Q)$: the cross constraints are vacuous. The inequalities cutting $\mathcal{R}(P \times Q)$ out of $\text{Box}(P) \times \text{Box}(Q)$ are therefore exactly the inequalities of P on the I -part together with those of Q on the J -part, which is to say $\mathcal{R}(P \times Q) = \mathcal{R}(P) \times \mathcal{R}(Q)$.

The defect. $|\mathcal{R}(P \times Q)| = |\mathcal{R}(P)| \cdot |\mathcal{R}(Q)| = (|P| + E(P)) \cdot (|Q| + E(Q))$, so

$$E(P \times Q) = (|P| + E(P))(|Q| + E(Q)) - |P||Q| = |P| \cdot E(Q) + |Q| \cdot E(P) + E(P) \cdot E(Q). \blacksquare$$

PROVED, MACHINE-CHECKED over every subset of $2 \times 2 \times 2$, $3 \times 3 \times 3$ and $2 \times 3 \times 4$ with the factors quantified over, the second factor one-dimensional; the exhaustive family covers two two-dimensional factors: **EXHAUSTIVE** over all 945 pairs (P, Q) with P a subset of a 3×2 box and Q a subset of a 2×2 , with 0 closure failures and 0 identity failures.

Defects therefore multiply as well as add across a product: a closed factor beside an open one already costs $|P| \cdot E(Q)$, and two open factors cost a cross term besides. Every term of the identity is non-negative and $|P|, |Q| \geq 1$, so with Theorem 3 a product is closed exactly when both its factors are.

§8 • Bands and the triangle

Two regions defined by absolute-value inequalities — a bounded difference and a triangle inequality — recur in indexing, and they behave differently. Both statements below are over **all** integers, not over a box.

Theorem 10 (a band is a sublattice). For every integer $k \geq 0$, $B_k = \{ (a, b) \in \mathbb{Z}^2 : |a - b| \leq k \}$ is closed under coordinatewise maximum and minimum.

Proof. Take $(a_1, b_1), (a_2, b_2) \in B_k$ and write $u = \max(a_1, a_2)$, $v = \max(b_1, b_2)$; choose $p, r \in \{1, 2\}$ with $u = a_p$ and $v = b_r$. Then

$$|u - v| = a_p - b_r \leq a_p - b_p \leq |a_p - b_p| \leq k, \text{ since } b_r \geq b_p; \quad v - u = b_r - a_p \leq b_r - a_r \leq |a_r - b_r| \leq k, \text{ since } a_p \geq a_r.$$

So $|u - v| \leq k$ and the join (u, v) lies in B_k . For the meet write $u' = \min(a_1, a_2)$ and $v' = \min(b_1, b_2)$, and choose p, r afresh with $u' = a_p$ and $v' = b_r$; then $u' - v' = a_p - b_r \leq a_r - b_r \leq |a_r - b_r| \leq k$ since $a_p \leq a_r$, and $v' - u' = b_r - a_p \leq b_p - a_p \leq |a_p - b_p| \leq k$ since $b_r \leq b_p$. ■ **PROVED, MACHINE-CHECKED** over all integers a_1, b_1, a_2, b_2, k simultaneously, the hypothesis checked satisfiable, and **EXHAUSTIVE** as a fixed point of $\mathcal{R}$ on a 6×6 box for $k = 0$ through 5.

Theorem 11 (the triangle region is join-closed and meet-broken). Let

$$\Delta = \{ (a, b, c) \in \mathbb{Z}^3 : |a - b| \leq c \leq a + b \}$$

— the triangle inequality, as it constrains a third quantity formed by combining two others. Then Δ is closed under coordinatewise maximum, and it is **not** closed under coordinatewise minimum.

Proof of join-closure. Take $(a_1, b_1, c_1), (a_2, b_2, c_2) \in \Delta$ and write $u = \max(a_1, a_2)$, $v = \max(b_1, b_2)$, $w = \max(c_1, c_2)$.

Upper bound. $w = c_k$ for some $k \in \{1, 2\}$, and $c_k \leq a_k + b_k \leq u + v$.

Lower bound. $u = a_k$ for some k , so $u - v \leq a_k - b_k \leq |a_k - b_k| \leq c_k \leq w$, the first step because $v \geq b_k$. Exchanging the roles of a and b gives $v - u \leq w$. Together, $|u - v| \leq w$.

Both inequalities hold, so the join lies in Δ . ■ **PROVED**, and **MACHINE-CHECKED** over all integers.

REFUTATION (the meet). Two witnesses, one for each inequality.

$(0,1,1) \wedge (1,0,1) = (0,0,1)$: here $|0 - 0| = 0 \leq 1$, but $1 \leq 0 + 0$ fails — the upper bound breaks, and the meet manufactures a combined quantity where neither part has any. $(4,0,4) \wedge (2,2,0) = (2,0,0)$: here $|2 - 0| = 2 \leq 0$ fails — the lower bound breaks.

Both source cells of each pair satisfy both inequalities. **REFUTATION**, re-verified, and confirmed by a solver over all integers, which reports the meet condition satisfiably violated.

The asymmetry has a reading. The maximum is the operation that can only move a coordinate toward the side each bound is slack on, so certainty survives upward and dies downward.

The same asymmetry with the constant replaced by a coordinate. Drop the upper bound and keep the lower: the region $C = \{ (a, b, c) \in \mathbb{Z}^3 : |a - b| \leq c \}$ is join-closed and is still not a sublattice.

Proof of join-closure. With u, v, w the coordinatewise maxima and $u = a_k$: $u - v \leq a_k - b_k \leq |a_k - b_k| \leq c_k \leq w$, and symmetrically $v - u \leq w$. ■ **PROVED, MACHINE-CHECKED** over all integers.

Meet-closure fails, and the failures are dense:

region $ a - b \leq c$ on	cells	unordered pairs	meet failures	join failures
$\{0, \dots, 8\}^3$	489	119,316	12,654	0
$\{0, \dots, 12\}^3$	1,469	1,078,246	113,568	0
$\{0, \dots, 16\}^3$	3,281	5,380,840	565,284	0

Table 2. Every unordered pair of cells of the region, at three caps. **EXHAUSTIVE**.

The difference between B_k and C is that k is a constant and c is a coordinate: a bound by a constant is monotone in both directions, a bound by a free coordinate is not. The consequence is that a region cut out by a triangle inequality cannot be carried exactly by an index of this kind. It can be carried as an envelope — the join-closed relaxation — and no further, at any number of coordinates.

§9 • The seed

A closed index is $\mathcal{R}$ of any of its generating sets — by Theorem 2, of any set that generates it as a lattice. The question is how small such a set can be, and it is answered on two families: one the paper's, one Czédli's.

Theorem 12 (the generation criterion). Let X be closed and $G \subseteq X$ non-empty. Then $\mathcal{R}(G) = X$ if and only if

- (a) $\pi_i(G) = \pi_i(X)$ for every coordinate i , and
- (b) given (a), $\phi_{ij}[G](a) = \phi_{ij}[X](a)$ for every $i \neq j$ and every $a \in A_j$ — (a) makes $\text{Box}(G) = \text{Box}(X)$, so $\phi[G]$ is defined on the same alphabets as $\phi[X]$.

Proof. Sufficiency. (a) gives $\text{Box}(G) = \text{Box}(X)$ and (b) gives the same constraints, so $\mathcal{R}(G)$ and $\mathcal{R}(X)$ are cut out of the same box by the same inequalities and are equal; and $\mathcal{R}(X) = X$ because X is closed (Theorem 3).

Necessity. Suppose $\mathcal{R}(G) = X$. By Lemma 3, $\text{Box}(\mathcal{R}(G)) = \text{Box}(G)$ and $\phi[\mathcal{R}(G)] = \phi[G]$. Substituting $\mathcal{R}(G) = X$ gives $\text{Box}(X) = \text{Box}(G)$, which is (a), and $\phi[X] = \phi[G]$, which is (b). ■ **PROVED**, and **MACHINE-CHECKED** over every subset of 3×3 and $2 \times 2 \times 2$, with both G and the closed set quantified over.

Theorem 13 (the seed is a minimum set cover). Let X be closed. Define a finite set of **elements**:

- slots** (i, v) for each coordinate i and each $v \in A_i$; **steps** (i, j, t) for each ordered pair $i \neq j$ and each $t \in A_j$ that is the least value of A_j at which ϕ_{ij} attains its value there.

Say a cell $g \in X$ **covers** the slot (i, v) if $g_i = v$, and covers the step (i, j, t) if $g_j \leq t$ and $g_i = \phi_{ij}(t)$. Then for $G \subseteq X$,

$$\mathcal{R}(G) = X \iff G \text{ covers every slot and every step.}$$

Hence $\text{seed}(X)$ is the optimum of a minimum set cover instance whose elements are the slots and steps and whose sets are the cells.

Proof. Condition (a) of Theorem 12 says precisely that for every i and every $v \in A_i$ some $g \in G$ has $g_i = v$ — that every slot is covered.

For (b): since $G \subseteq X$, monotonicity of ϕ in its set (the argument of Theorem 1(ii)) gives $\phi_{ij}[G](a) \leq \phi_{ij}[X](a)$ wherever $\phi[G]$ is defined, so (b) is the assertion that no inequality is strict.

Suppose every step is covered. Let $a \in A_j$ and let $t \leq a$ be the largest step for the pair (i, j) with $t \leq a$; such a t exists because the least element of A_j is a step, ϕ_{ij} being defined there. Then $\phi_{ij}[X](a) = \phi_{ij}[X](t)$: the value $\phi_{ij}[X](a)$ is first attained at some step $t'' \leq a$, and $t'' \leq t$ because t is the largest step at or below a ; so by Lemma 1, $\phi_{ij}[X](t'') \leq \phi_{ij}[X](t) \leq \phi_{ij}[X](a) = \phi_{ij}[X](t'')$, and the three are equal. Some $g \in G$ covers (i, j, t) , so $g_j \leq t \leq a$ and $g_i = \phi_{ij}[X](t)$. Hence $\phi_{ij}[G](a) \geq g_i = \phi_{ij}[X](t) = \phi_{ij}[X](a)$, and with the reverse inequality above, (b) holds at a .

Conversely suppose (b) holds and let (i, j, t) be a step. Then $\phi_{ij}[G](t) = \phi_{ij}[X](t)$, and the maximum defining $\phi_{ij}[G](t)$ is attained at some $g \in G$ with $g_j \leq t$ and $g_i = \phi_{ij}[X](t)$ — which is the covering condition for that step. ■ **PROVED**, and **EXHAUSTIVE**: over all 5,111 pairs (G, X) with X a closed subset of a 3×3 or a $2 \times 2 \times 2$ box and G any non-empty subset of X , the covering test and direct closure agree in every case, 0 disagreements.

Minimum set cover is NP-complete — **Karp (1972)**, CITED. Theorem 13 places seed inside that problem and so bounds its difficulty from above; it does not show seed hard, and no hardness is claimed. No general formula is claimed either. There is one on each of two families.

The lower bound, and what the Carathéodory number does not give

Proposition 5 (the alphabet bound). For any closed X , $\text{seed}(X) \geq \max_i |A_i|$.

Proof. A single cell g has one value at each coordinate, so for a fixed i it covers exactly one slot (i, g_i) . The $|A_i|$ slots at coordinate i therefore need at least $|A_i|$ distinct cells. ■ **PROVED**, and **EXHAUSTIVE** over all 219 closed subsets of a 3×3 and a $2 \times 2 \times 2$ box, 0 violations.

The natural guess for a second lower bound comes from convexity. In a convexity space the **Carathéodory number** is the least κ such that any point of the hull of a set lies in the hull of at most κ of its members — the name is from the theorem of Carathéodory (1911) on convex hulls in Euclidean space. For a semilattice with its subsemilattices as the convex sets the Carathéodory number is the **breadth** (Jamison-Waldner 1982; van de Vel 1993), and the breadth of a product of d chains is d ; the Carathéodory, Helly and Radon numbers of the sublattice convexities of product spaces are determined in Queyranne and Tardella (2017) — CITED. It is tempting to read that as $\text{seed} \geq d$. It is not that.

REFUTATION (seed $\geq d$). The two-element chain $\{(0,0,0), (1,1,1)\}$ is a closed subset of $\{0,1\}^3$, with $d = 3$, and it is generated by both of its cells: $\text{seed} = 2 < d$. **REFUTATION**, re-verified.

The Carathéodory number bounds how many generators are needed to reach *one point of the hull*; the seed is how many are needed to reach *every point* at once, and the two are different quantities. What the breadth is the breadth *of* also matters: the breadth of the chain above is 1, not 3. Proposition 5 is the bound that survives, and on the witness it gives $\text{seed} \geq 2$, which is tight.

The ordered simplex

For $d \geq 2$ and $c \geq 2$ let $D(d, c) = \{x \in \{0, \dots, c-1\}^d : x_i \geq x_{i+1} \text{ for } 1 \leq i < d\}$, the non-increasing tuples, $C(d + c - 1, d)$ cells. In the product order $D(d, c)$ is a sublattice of Box but not a down-set of it: $(1,1) \in D(2,2)$ while $(0,1) \leq (1,1)$ is not. As a lattice it is $L(d, c-1)$, the Young diagrams that fit in a $d \times (c-1)$ box ordered by inclusion (Stanley 1980), a tuple's coordinates being the row lengths.

Theorem 14 (the simplex seed law). For $d \geq 2$ and $c \geq 2$, $D(d, c)$ is closed, and $\text{seed}(D(d, c)) = d + c - 1$. The minimum seed is unique: it is the c constant cells $(t, t, \dots, t)$ for $t = 0, \dots, c-1$ together with the $d - 1$ cells $(c-1)^k 0^{d-k}$ for $k = 1, \dots, d-1$, and every one of them is forced.

Proof. *The boundary functions.* For $i < j$ and $a \in \{0, \dots, c-1\}$, the cell with the first $j - 1$ coordinates at $c-1$ and the rest at a is non-increasing, has j -coordinate a and i -coordinate $c-1$, so $\varphi_{ij}(a) = c-1$: the constraint is vacuous. For $i > j$, any non-increasing y with $y_j \leq a$ has $y_i \leq y_j \leq a$, and the constant cell $(a, \dots, a)$ attains it, so $\varphi_{ij}(a) = a$. Hence $\mathcal{R}(D) = \{x : x_i \leq x_j \text{ whenever } i > j\} = D$, and D is closed.

Forced cells. By Theorem 13, an element covered by exactly one cell forces that cell into every seed. For $i < j$ the only step is at $t = 0$ — φ_{ij} being constant — and a cell covering $(i, j, 0)$ has $g_j = 0$ and $g_i = c-1$; being non-increasing it then has $g_1 = \dots = g_i = c-1$ and $g_j = \dots = 0$ from coordinate j to the last, so it is $(c-1)^i 0^{d-i}$ for some $i \leq p < j$. Taking $j = i + 1$ leaves exactly one choice, $p = i$. So each of the $d - 1$ cells $(c-1)^k 0^{d-k}$, $k = 1, \dots, d-1$, is forced, by the step $(k, k+1, 0)$. For $i > j$ every t is a step, since $\varphi_{ij}(t) = t$ is strictly increasing; a cell covering $(d, 1, t)$ has $g_1 \leq t$ and its last coordinate equal to t , and being non-increasing has every coordinate equal to t . So each of the c constant cells is forced, by the step $(d, 1, t)$. The $d + c - 1$ cells named are pairwise distinct: the constant cells differ from one another, and $(c-1)^k 0^{d-k}$ with $1 \leq k \leq d-1$ is non-constant because $c - 1 > 0$. So $\text{seed} \geq d + c - 1$.

They generate. Slots: the slot (i, v) is covered by the constant cell $(v, \dots, v)$. Steps with $i < j$: the step $(i, j, 0)$ is covered by $(c-1)^j 0^{d-i}$, which is in the list because $i \leq d-1$. Steps with $i > j$: the step (i, j, t) is covered by $(t, \dots, t)$, which has $g_j = t \leq t$ and $g_i = t = \varphi_{ij}(t)$. Every element is covered, so by Theorem 13 the $d + c - 1$ cells generate, and $\text{seed} = d + c - 1$. Uniqueness follows because all $d + c - 1$ are forced and they suffice. ■ **PROVED**, and **EXHAUSTIVE** at nine instances — $D(2,2)$, $D(2,3)$, $D(3,2)$, $D(3,3)$, $D(2,4)$, $D(3,4)$, $D(4,3)$, $D(4,4)$, $D(5,3)$ — where the forced cells are computed from the cover instance and found to be exactly the predicted set, the minimum cover is computed by branch and bound, and for the smaller instances the seed is confirmed a second time by direct closure over every subset. Theorem 14 is not a case of Czédli’s theorem below, which concerns direct powers, and no statement of it in the literature is known.

The full box

The full box behaves differently, and the difference is where a plausible linear law fails.

Theorem 15 (the box seed law — Czédli). Let $c \geq 2$, $d \geq 2$, and let $Q(c, d) = \{0, \dots, c-1\}^d$. Put $m(d) := \min \{ m : C(m, \lfloor m/2 \rfloor) \geq d \}$. Then

$$\left| \begin{array}{l} \text{seed}(Q(c, d)) = c - 2 + m(d). \end{array} \right.$$

Equivalently, for $k \geq c$: some k cells generate $Q(c, d)$ for d as large as $C(k - c + 2, \lfloor (k - c + 2)/2 \rfloor)$, and no k cells generate it for any larger d . (For $k < c$ no k cells generate it, by Proposition 5.)

CITED. By the gloss on D8, $\text{seed}(Q(c, d))$ is the least size of a generating set of the lattice $Q(c, d)$, the d -th direct power of a c -element chain. At $c = 2$ the theorem is Theorem 2.1 of **Czédli (2023a)**: the Boolean lattice with d atoms has an at most k -element generating set if and only if $d \leq C(k, \lfloor k/2 \rfloor)$, proved there from Sperner’s theorem by the antichain argument below. For every c it is a case of Theorem 2.4 of **Czédli (2023b)**, which makes $k \mapsto$ (the least generating size of D^k) the left adjoint of $n \mapsto S(J(D), n)$ for any finite distributive lattice D with poset of join-irreducibles $J(D)$, $S(U, n)$ being the largest number of pairwise unrelated copies of U among the subsets of an n -element set; Observation 3.1 there gives $S(U, n) = C(n - p, \lfloor (n - p)/2 \rfloor)$ for a bounded U that embeds in the subsets of a p -element set and no smaller, which for a chain is the theorem of **Griggs, Stahl and Trotter (1984)**. A c -element chain has a $(c-1)$ -element chain of join-irreducibles, embedding at $p = c - 2$, and the adjoint is then $c - 2 + m(d)$. The proof below is this paper’s own, from Theorem 13 and the set-pair inequality; it is kept because it is short, because it is stated in the terms the refutation that follows needs, and because it reaches the value by a different route.

Proof. Present a candidate G of k cells as a $k \times d$ matrix whose rows are the cells; column j is then a vector $v_j \in \{0, \dots, c-1\}^k$ recording what the j -th coordinate does across G .

The covering conditions, in matrix terms. Every boundary function of the full box is constant at $c - 1$, so by Theorem 13 the elements are: the slots (j, v) , which demand that each column v_j be surjective onto $\{0, \dots, c-1\}$; and, for each ordered pair (i, j) , the single step at $t = 0$, which demands a row r with $v_j[r] = 0$ and $v_i[r] = c-1$.

Upper bound. Suppose G , of size k , generates $Q(c, d)$. For each column j set $P_j = \{ r : v_j[r] = c-1 \}$ and $Z_j = \{ r : v_j[r] = 0 \}$. These are disjoint, since $c \geq 2$. Surjectivity forces P_j and Z_j non-empty and forces at least one further row for each of the $c - 2$ intermediate values, so

$$\left| \begin{array}{l} |P_j| + |Z_j| \leq k - (c - 2) =: m, \end{array} \right.$$

and $m \geq 2$, since $k \geq c$ by Proposition 5, the largest alphabet of $Q(c, d)$ being c .

The step condition for the ordered pair (i, j) says exactly $P_i \cap Z_j \neq \emptyset$, for every $i \neq j$. The family (P_j, Z_j) , $j = 1, \dots, d$, therefore satisfies the hypotheses of the **set-pair inequality of Bollobás (1965)** — CITED — which gives

$$\left| \begin{array}{l} \sum_j 1 / C(|P_j| + |Z_j|, |P_j|) \leq 1, \text{ the sum over } j = 1, \dots, d. \end{array} \right.$$

Every term is at least $1 / C(m, \lfloor m/2 \rfloor)$, because $C(n, a) \leq C(m, \lfloor m/2 \rfloor)$ whenever $n \leq m$, the central binomial coefficient being the largest of its row and increasing in the row. Hence $d / C(m, \lfloor m/2 \rfloor) \leq 1$, that is $d \leq C(k - c + 2, \lfloor (k - c + 2)/2 \rfloor)$.

Lower bound, by construction. Let $m = k - c + 2$ and let $\mathcal{A}$ be the family of all $A \subseteq \{1, \dots, m\}$ with $|A| = \lfloor m/2 \rfloor$, of size $C(m, \lfloor m/2 \rfloor)$. Index the rows by $\{1, \dots, m\}$ together with $c - 2$ extra rows, one labelled by each $t = 1, \dots, c - 2$. For each $A \in \mathcal{A}$ define a column by

$$v[r] = c - 1 \text{ for } r \in A; v[r] = 0 \text{ for } r \in \{1, \dots, m\} \setminus A; v = t \text{ on the extra row labelled } t, \text{ for } t = 1, \dots, c - 2.$$

Each such column is surjective onto $\{0, \dots, c - 1\}$, since A and its complement in $\{1, \dots, m\}$ are both non-empty for $m \geq 2$. For two distinct $A, A' \in \mathcal{A}$, equal cardinality forces $A \not\subseteq A'$, so some row lies in $A \setminus A'$, where the first column reads $c - 1$ and the second reads 0; symmetrically for the other order. Every ordered pair of columns therefore satisfies its step condition. Choosing any $d \leq C(m, \lfloor m/2 \rfloor)$ of these columns gives a matrix of $m + c - 2 = k$ rows whose set of rows generates $Q(c, d)$, so $\text{seed}(Q(c, d)) \leq k$.

Conclusion. The least k with $C(k - c + 2, \lfloor (k - c + 2)/2 \rfloor) \geq d$ is $k = c - 2 + m(d)$, since $C(m, \lfloor m/2 \rfloor)$ is increasing in m . ■ **PROVED** here and **CITED** (Czédli 2023a at $c = 2$; Czédli 2023b with Griggs, Stahl and Trotter 1984 for every c), and **EXHAUSTIVE** on two fronts: the seed is confirmed by direct closure — every subset of size below the stated value fails and one of that size generates — at $2^2, 3^2, 2^3, 3^3, 2^4, 2^5, 4^2$ and 5^2 , with the generating set printed in each case; and the clique count $C(k - c + 2, \lfloor \cdot \rfloor)$ is confirmed by exact maximum-clique search on the compatibility graph at $c = 2$ for $k = 2, \dots, 7$ (largest $d = 2, 3, 6, 10, 20, 35$), at $c = 3$ for $k = 3, \dots, 6$ ($2, 3, 6, 10$) and at $c = 4$ for $k = 4, \dots, 6$ ($2, 3, 6$) — thirteen counts in all, each equal to $C(k - c + 2, \lfloor \cdot \rfloor)$.

At $c = 2$ the construction and the bound are Sperner's theorem: the columns correspond to subsets of the row set, the step condition says two subsets are incomparable, a generating set is an antichain, and the largest antichain in a Boolean lattice has $C(k, \lfloor k/2 \rfloor)$ members — **Sperner (1928)**, **CITED**, by way of the LYM inequality (Lubell 1966), of which Bollobás's inequality is the generalisation used above. That is the argument of Czédli (2023a). One value is worth printing. For the five-element chain at $d = 2023$ the law gives $3 + m(2023) = 17$, since $C(13, 6) = 1,716 < 2,023 \leq 3,432 = C(14, 7)$; table (4.30) of Czédli (2023b) prints 18 at that entry, while Theorem 2.4 there with $p = 3$ and the adjoint of the central binomial coefficient at 14 gives 17, as the law does. The check confirms the shift $c - 2$, and not $c - 1$, by direct closure at 5^2 : five cells generate it and no four do.

REFUTATION (the linear law). $\text{seed}(Q(c, d)) = d + c - 2$ holds for small d — it agrees with Theorem 15 at $d = 2, 3, 4$ for every c , since $m(2) = 2, m(3) = 3$ and $m(4) = 4$ — and **fails from $d = 5$** : the box $\{0, 1\}^5$ is generated by **four** cells, where $d + c - 2 = 5$. The four are $(0, 0, 0, 1, 1), (0, 1, 1, 0, 0), (1, 0, 1, 0, 1), (1, 1, 0, 1, 0)$ — the five columns being five of the six two-element subsets of a four-element row set, an antichain, as Theorem 15 requires. **REFUTATION**, re-verified by direct closure: no three cells generate $\{0, 1\}^5$.

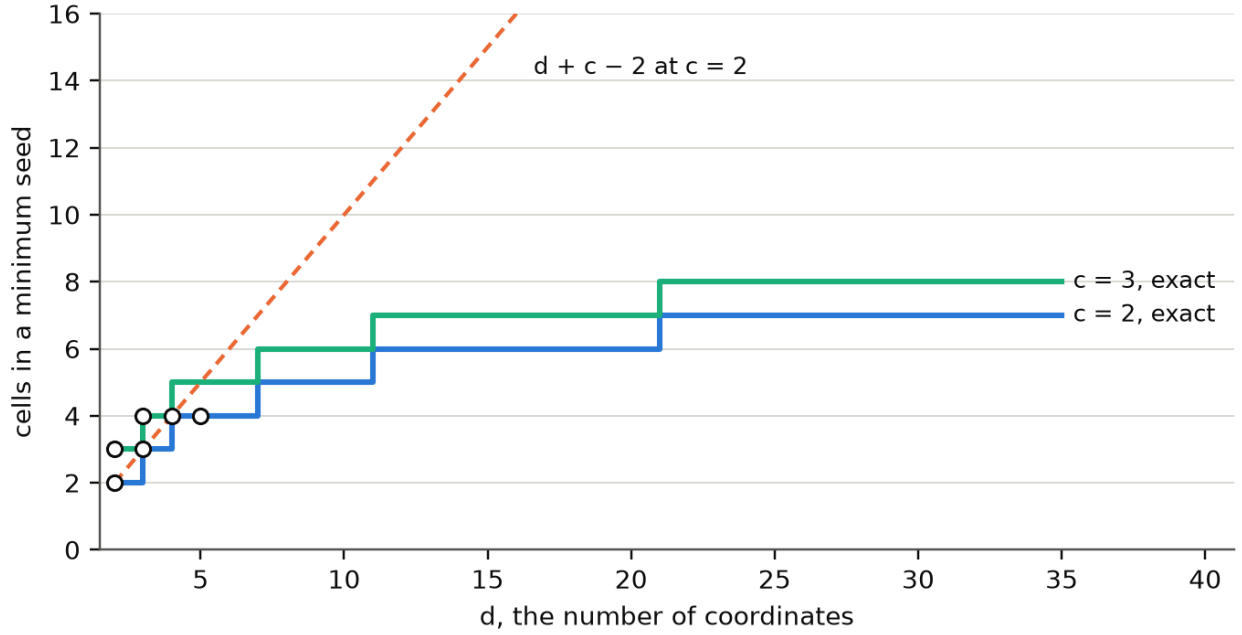


Figure 5. The seed of the full box against the number of coordinates, at alphabet sizes $c = 2$ and $c = 3$. The exact law $c - 2 + m(d)$ of Theorem 15 (solid, stepped) grows like $\log d$; the linear guess $d + c - 2$ (dashed, drawn at $c = 2$) agrees at $d = 2, 3, 4$ and diverges from $d = 5$ on. Circles mark the six instances at $c = 2$ and $c = 3$ confirmed by direct closure; a seventh, 4^2 with seed 4, and an eighth, 5^2 with seed 5, are not plotted.

The two laws pull in opposite directions and both are exact. The ordered simplex on c values in d coordinates needs $d + c - 1$ cells, linear in the dimension; a full box of the same alphabet needs $c - 2 + m(d)$, which is logarithmic in it. The difference is the number of boundary steps each family carries: the simplex has a step at every value of every ordered pair, the full box has one per pair and a trivial one at that.

A note on scope. The electron-configuration lattice is a closed index of this kind, and its own seed is the subject of a companion paper; it is not developed here.

§10 · Verification record

Every number printed above is recomputed by the machine checks, or marked CITED. The checks are **53 machine-checked obligations**, **47 exhaustive families**, **6 refutations** and **8 cited results**, discharged behind **20 guards**, with 0 failures. Guards are reported first and no obligation is reported if a guard fails. The two tables below — §2 to §5, then §6 to §9 — are one row per result; the boxes named in them are the 53 machine-checked obligations, and the families named after them are the 47 exhaustive ones.

object	PROVED	MACHINE-CHECKED (boxes)	EXHAUSTIVE (family)	other
Lemma 1, φ total and isotone	✓	—	—	—
Lemma 2, witness form	✓	—	—	used as the encoding of D5
Lemma 3, $\mathcal{R}$ fixes box and boundary	✓	—	—	—
Proposition 1, defect range	✓	—	witness at 2^2	—

object	PROVED	MACHINE-CHECKED (boxes)	EXHAUSTIVE (family)	other
Theorem 1 (i) extensive	✓	3×3, 4×4, 2×2×2, 3×3×3	74,569 subsets, seven boxes	—
Theorem 1 (ii) monotone	✓	3×3, 4×4, 2×2×2, 3×3×3	26,206 nested pairs, four boxes	—
Theorem 1 (iii) idempotent	✓	3×3, 4×4, 2×2×2, 3×3×3	74,569 subsets, seven boxes	—
Lemma 4 , $\mathcal{R}(X)$ a sublattice	✓	3×3, 4×4, 2×2×2, 3×3×3	—	—
Lemma 5, four witnesses (d = 2)	✓	—	815,072 cells of $\mathcal{R}(X)$	—
Theorem 2 , $\mathcal{R}(X) = \langle X \rangle$	✓ at d = 2; CITED lift	3×3, 4×4, 2×2×2, 3×3×3, 2×2×2×2	74,569 subsets (boxes ≤ 12 cells)	CITED Baker & Pixley 1975
Theorem 3 , closed $\iff E = 0$	✓	3×3, 4×4, 2×2×2, 3×3×3	74,569 subsets, seven boxes	—
Theorem 4 , the least bound system	✓	3×3, 2×2×2	—	—
Theorem 5 , bands at d = 2	✓	3×3, 4×4, 3×5	3,713 closed sets	—
the nine-cell staircase	—	—	9 cells, E = 0, extremes read back	—
Theorem 6 , Moore family	✓	3×3, 4×4, 2×2×2, 3×3×3	seven ambients (Table 1)	—
union of closed sets	—	—	—	REFUTATION , 2 × 2 witness
Proposition 2, separation	✓	—	seven ambients, every ordered pair	—
Theorem 7 , $E(\text{Cl}(U)) = 2^n - \text{Cl}(U) $	✓	—	seven ambients (Table 1), $ \mathcal{R}(\text{Cl}(U)) $ computed directly	—

object	PROVED	MACHINE-CHECKED (boxes)	EXHAUSTIVE (family)	other
Theorem 8 , projections	✓	2×2×2, 3×3×3, 2×3×4, two pairs each	438 projections	—
the converse of Theorem 8	—	—	117 subsets of 2×2×2, 52 open	REFUTATION , six-cell witness
Theorem 9 , graph criterion	✓	3×3, 2×2×2	—	—
Corollary 1, adjunction never repairs	✓	—	—	—
Proposition 3 , the difference	✓	3×3, 4×4, 3×3×3	—	—
Remark, the difference is an interval map	✓	—	—	—
Proposition 4 , product rule	✓	2×2×2, 3×3×3, 2×3×4	945 pairs	—
Theorem 10 , band $ a-b \leq k$	✓	all integers	6 × 6, k = 0...5	—

object	PROVED	MACHINE-CHECKED (boxes)	EXHAUSTIVE (family)	other
Theorem 11 , triangle join-closed	✓	all integers	—	—
the triangle's meets, by solver	—	—	—	REFUTATION : sat over all integers
the triangle's meets, by witness	—	—	—	REFUTATION , two explicit witnesses
region $ a-b \leq c$	✓ (join)	all integers (join)	three caps (Table 2)	meets fail, counted
Theorem 12 , generation criterion	✓	3×3, 2×2×2	—	—
Theorem 13 , seed = set cover	✓	—	5,111 pairs (G, X)	CITED Karp 1972: set cover is NP-complete
Proposition 5 , alphabet bound	✓	—	219 closed sets	—
seed $\geq d$	—	—	—	REFUTATION , three-coordinate chain
Theorem 14 (simplex), seed($D(d,c)$) = $d+c-1$	✓	—	nine simplices	—
Theorem 15 , seed(c^d) = $c-2+m(d)$	✓ (second proof)	—	eight boxes + 13 clique counts	CITED Czédli 2023a ($c = 2$), Czédli 2023b with Griggs, Stahl & Trotter 1984 (all c); Bollobás 1965, Sperner 1928 in the proof
seed(c^d) = $d + c - 2$	—	—	—	REFUTATION , $\{0,1\}^5$ in four cells

The exhausted families, named. EXHAUSTIVE means a decision procedure visited every case in a stated finite family, so the families are named rather than left as counts.

- **74,569 subsets** — every non-empty subset of 2×2 , 3×2 , $2 \times 2 \times 2$, 3×3 , 4×3 , $2 \times 2 \times 3$ and $2 \times 2 \times 2 \times 2$. The $\mathcal{R}(X)$ = $\langle X \rangle$ column of that sweep is taken over the boxes of at most twelve cells.
- **26,206 nested pairs** — every pair $X \subseteq Y$ inside 2×2 , 3×2 , $2 \times 2 \times 2$ and 3×3 .
- **815,072 cells** — every cell of $\mathcal{R}(X)$, over every subset X of a 3×3 and a 4×4 box, with the four witnesses of Lemma 5 constructed and the identity $(y \vee u) \wedge (z \vee w) = x$ checked at each.
- **Seven ambients** (Table 1) — 2×2 , 3×2 , $2 \times 2 \times 2$, 3×3 , 4×3 , $2 \times 2 \times 3$, $2 \times 2 \times 2 \times 2$: every closed subset enumerated, every unordered pair of distinct members tested under $\cap$ and $\cup$, every ordered pair of cells tested for separation, the meet-irreducibles counted, and the family's own closure and defect computed.
- **3,713 closed sets** — every closed subset of a 4×4 and a 3×5 box, each tested against its own fibre extremes.
- **438 projections** — every closed subset of $2 \times 2 \times 2$ projected onto each of the six proper coordinate sets.
- **117 subsets** — every subset of $2 \times 2 \times 2$ with at least two cells whose three two-fold projections are all closed.
- **945 pairs** — every P in a 3×2 box against every Q in a 2×2 .
- **Three caps** (Table 2) — every unordered pair of cells of $\{(a,b,c) : |a-b| \leq c\}$ at caps 8, 12 and 16.
- **5,111 pairs (G, X)** — every closed subset X of a 3×3 or $2 \times 2 \times 2$ box against every non-empty $G \subseteq X$.
- **219 closed sets** — every closed subset of a 3×3 or a $2 \times 2 \times 2$ box, each with its exact minimum cover computed.
- **Nine simplices** — $D(2,2)$, $D(2,3)$, $D(3,2)$, $D(3,3)$, $D(2,4)$, $D(3,4)$, $D(4,3)$, $D(4,4)$, $D(5,3)$.

- **Eight boxes + 13 clique counts** — full boxes $2^2, 3^2, 2^3, 3^3, 2^4, 2^5, 4^2, 5^2$ by direct closure, and the value of the law at (5, 2023); maximum cliques at $c = 2, k = 2 \dots 7$, at $c = 3, k = 3 \dots 6$, and at $c = 4, k = 4 \dots 6$.
- **The four worked defects of §2** — 365 calendar cells; 90 periodic-table cells at each of two placements of helium; 56 ordered triples.
- **Figure 1's index** — the five cells of §2, closed to 9 cells, $E = 4$.

What the machine checks but that enumeration cannot. Each obligation is a quantified formula whose variables range over *every* subset of the named box — and, where the hull appears, over every closed superset as well. The solver returns *unsat* on the negation. Over a $3 \times 3 \times 3$ box that is $2^{27} = 134,217,728$ subsets, well past the enumeration frontier; over a $2 \times 2 \times 2 \times 2$ box, with a second subset quantified over beside the first, it is $2^{16} \times 2^{16}$ assignments. Two encodings make it decidable. The maximum and ϕ are eliminated by Lemma 2 — membership is the existence of a witness. And the hull is encoded without a least fixed point, as the intersection of all closed supersets, so “ $\mathcal{R}(X) \subseteq \langle X \rangle$ ” becomes “ $\mathcal{R}(X) \subseteq S$ for every closed $S \supseteq X$ ”, which is first-order. The three integer obligations — Theorems 10 and 11 and the join-closure of $|a - b| \leq c$ — and the one integer refutation, the triangle's meet, quantify over all integers and are not restricted to a box at all.

The two guards, and an unsat means nothing without them. *Encoding fidelity:* the predicate the solver reasons about must be the operator. Three predicates are checked concretely against independent implementations. The staircase predicate is evaluated on **2,919 cells** of the declared boxes against the operator under test, with 0 disagreements, and a fresh independent implementation of D1–D5 is compared against the same operator on **400 random instances** over six box shapes, with 0 disagreements. The closure predicate of D6 — which every fixed-point hypothesis and the Moore-family, projection, graph and band obligations rest on — is evaluated on **300 random sets** over five box shapes against a direct test of closure, with 0 disagreements. And the hull encoding, the intersection of all closed supersets, is computed by brute force on **60 instances** and compared against the generated sublattice, with 0 disagreements. Two negative controls must fail: a deliberately wrong reference — the join-closure in place of the staircase — disagrees on 24 of 50 instances, and the witness form *without* the box conjunct, which is a different operator, disagrees on 73 of 2,851 cells. A check that merely restated the formula would confirm a transcription and not an encoding. The remaining predicates — the band form of Theorem 5, the graph closure and homomorphism of Theorem 9, the chain and isotonicity of Proposition 3, the factor product of Proposition 4, the alphabet and reach conditions of Theorem 12 — are transcriptions of their definitions and are not separately guarded; each is built from the guarded closure predicate and from the coordinatewise operations that the exhaustive checks also use. *Non-vacuity:* an unsatisfiable hypothesis makes an implication vacuously true, so each of the five non-trivial hypothesis shapes used above — $X \subseteq S$ with S closed and proper; two distinct proper fixed points; X strictly inside Y ; G strictly inside a closed X with $\mathcal{R}(G) = X$; and Theorem 4's, a non-empty X cut out of its box by an isotone integer system — is checked satisfiable, and satisfiable non-trivially: the first four at each of 3×3 , $2 \times 2 \times 2$ and $3 \times 3 \times 3$, the fifth at 3×3 and $2 \times 2 \times 2$. Fourteen such checks, all passed. The hypotheses of Theorem 9 (h bounded into a three-element chain), Proposition 4 (two non-empty factors) and Theorem 5 (isotone $L \leq U$) are satisfied by explicit instances and are not separately checked.

The negative controls on the whole run. Three deliberately false claims are submitted and must be reported as refuted: $\mathcal{R}(X) \subseteq X$, which the solver refutes with a counterexample over 3×3 ; that the family of closed sets is union-closed, refuted by 7 unordered pairs of distinct closed subsets of a 2×2 box whose union escapes; and $\text{seed}(c^d) = d + c - 2$, refuted by direct closure at $(c, d) = (2, 5)$, where every three-cell subset of $\{0,1\}^5$ fails to generate and a four-cell one is found. Each is refuted by a decision procedure that could have returned the other answer, so a clean run is evidence and not a restatement.

What is not machine-checked, and why. Theorem 2's lift above $d = 2$ is cited, not checked in general: it is a theorem about a variety of algebras and is not of the shape a finite-box obligation has. The machine checks cover it over the five named boxes and no further, and §0 says so. Lemmas 1, 2, 3 and Corollary 1 carry written proofs only. Proposition 1, Proposition 2, Theorems 7, 14 and 15 carry written proofs and are confirmed over the ex-

haustive families named above rather than by a solver — Theorems 14 and 15 quantify over d and c , which no finite box bounds, and Theorem 7 quantifies over an ambient; Theorem 15 is besides cited in full. Theorem 13’s complexity clause — that minimum set cover is NP-complete — is cited and not re-derived. The two sampled soundness guards and the sampled closure guard are the only SAMPLED figures anywhere in this paper, and none is a result.

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